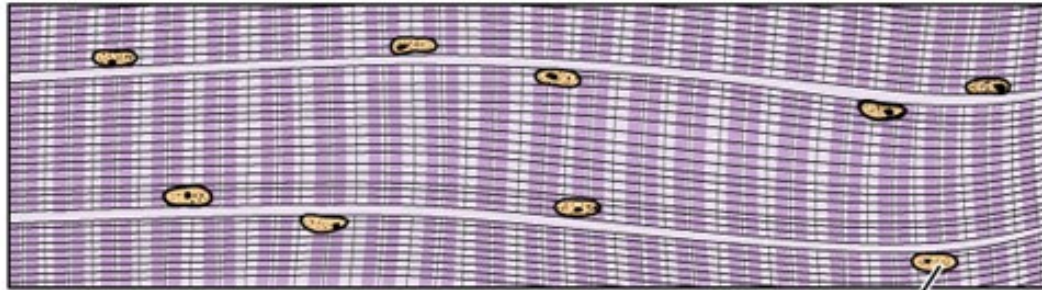


# Muscular Tissue



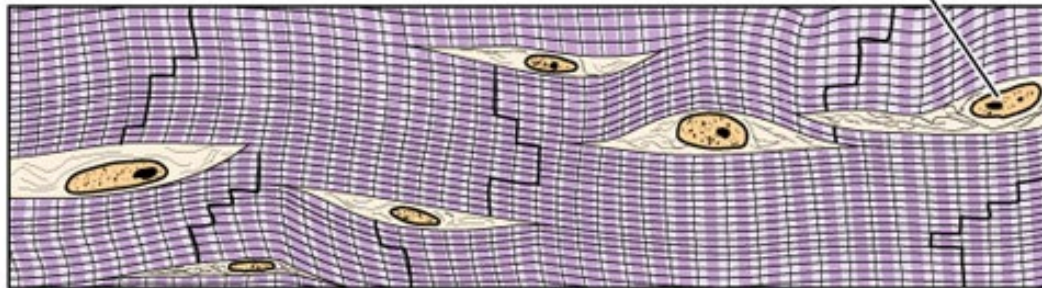
# Muscle types

## Skeletal muscle

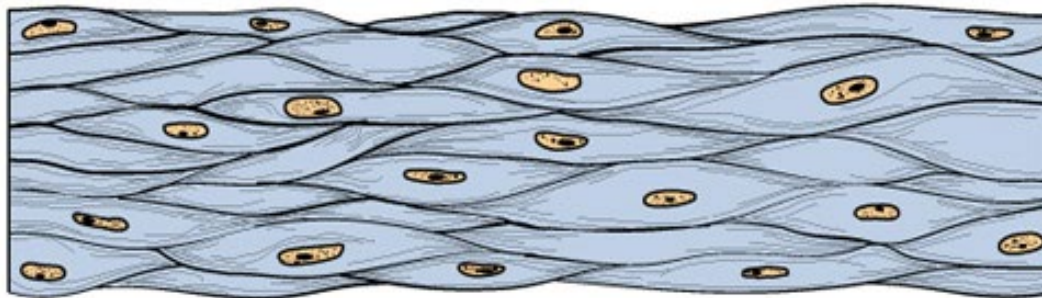


Nuclei

## Cardiac muscle

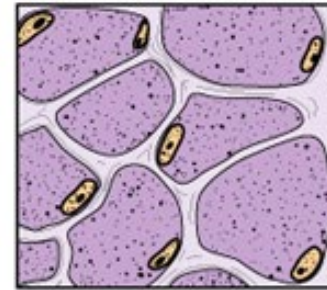


## Smooth muscle

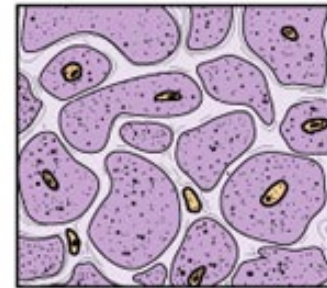


Intercalated disks

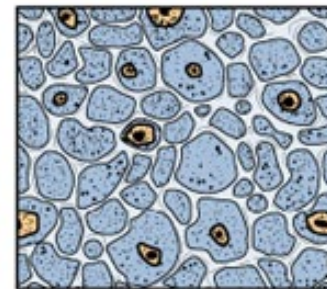
## Cross sections



- striation/voluntary
- Long cylindrical
- Multinucleated
- Myofibril



- Striation/invol.
- Cylindrical
- 1-2 nuclei
- Intercalated disc



- No striation/invol.
- Thin, long
- Spindle shape
- 1 nucleus
- Wall of tubular st.

**#6** skeletal muscle, HE, human, x400

– skeletal muscle(longitudinally, cross)

**#8** skeletal muscle, NADH-TR, rat

– type of skeletal muscle fiber

**#57** Uterus, monkey, HE, Smooth muscle

– smooth muscle

**#58** uterus, HE, human, X400

– smooth muscle (longitudinally, cross)

**#73** Heart, HE, monkey, x400

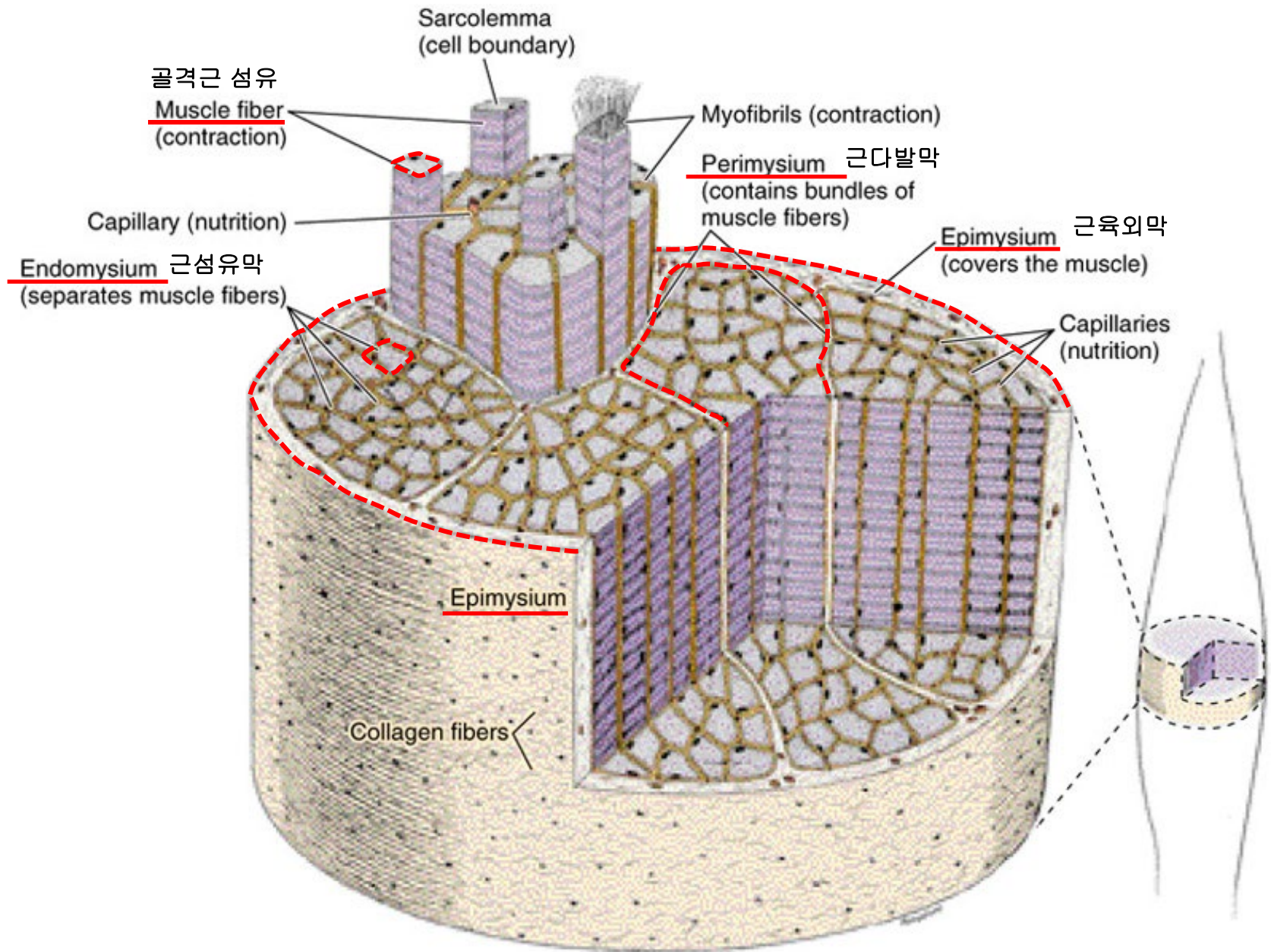
– cardiac muscle (longitudinally, cross)

**#19** Heart, HE, dog, x400 – Purkinje fiber

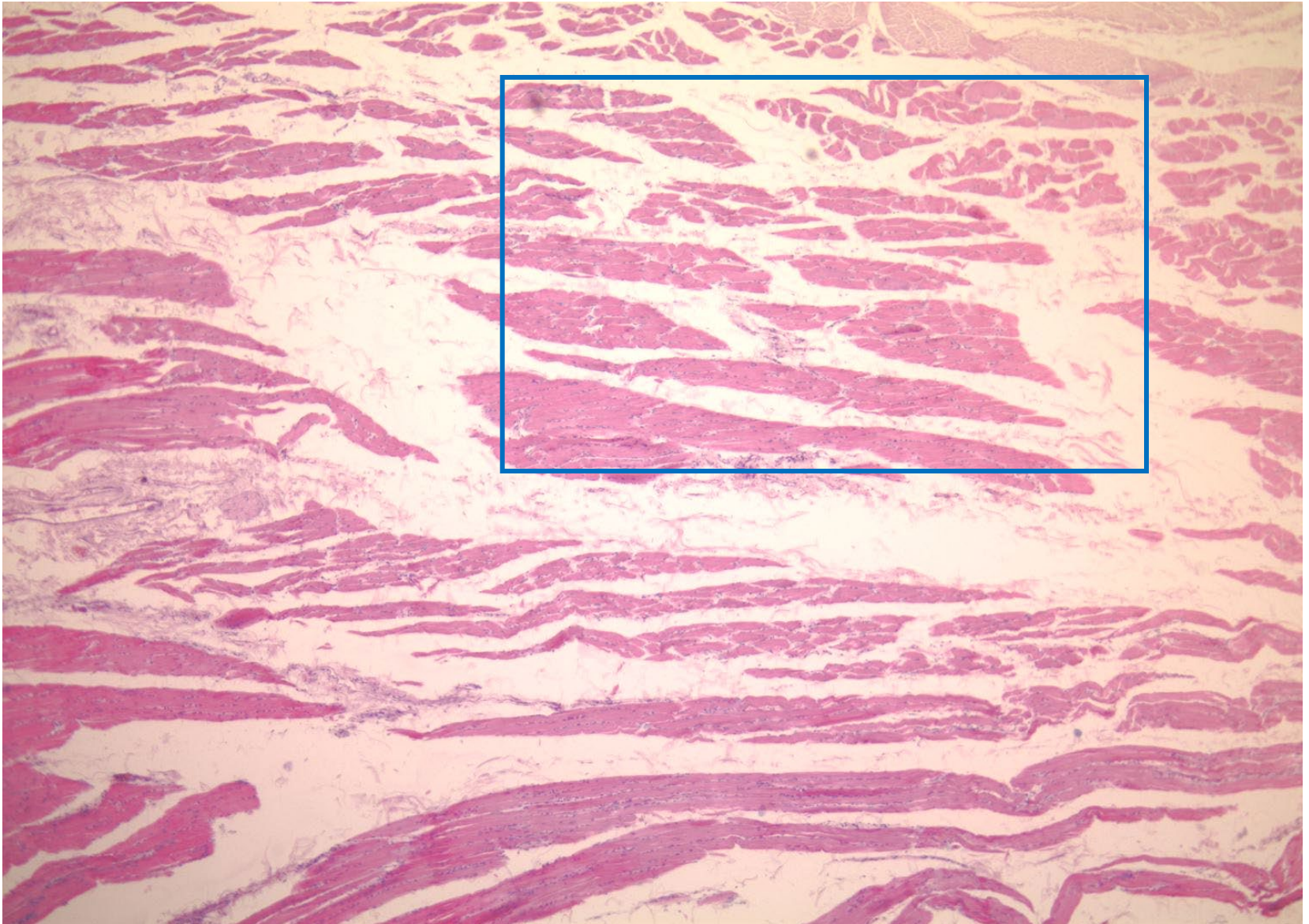
**#7** Skeletal muscle, modified Gomori trichrome, rat

– Neuromuscular spindle

# Skeletal muscle

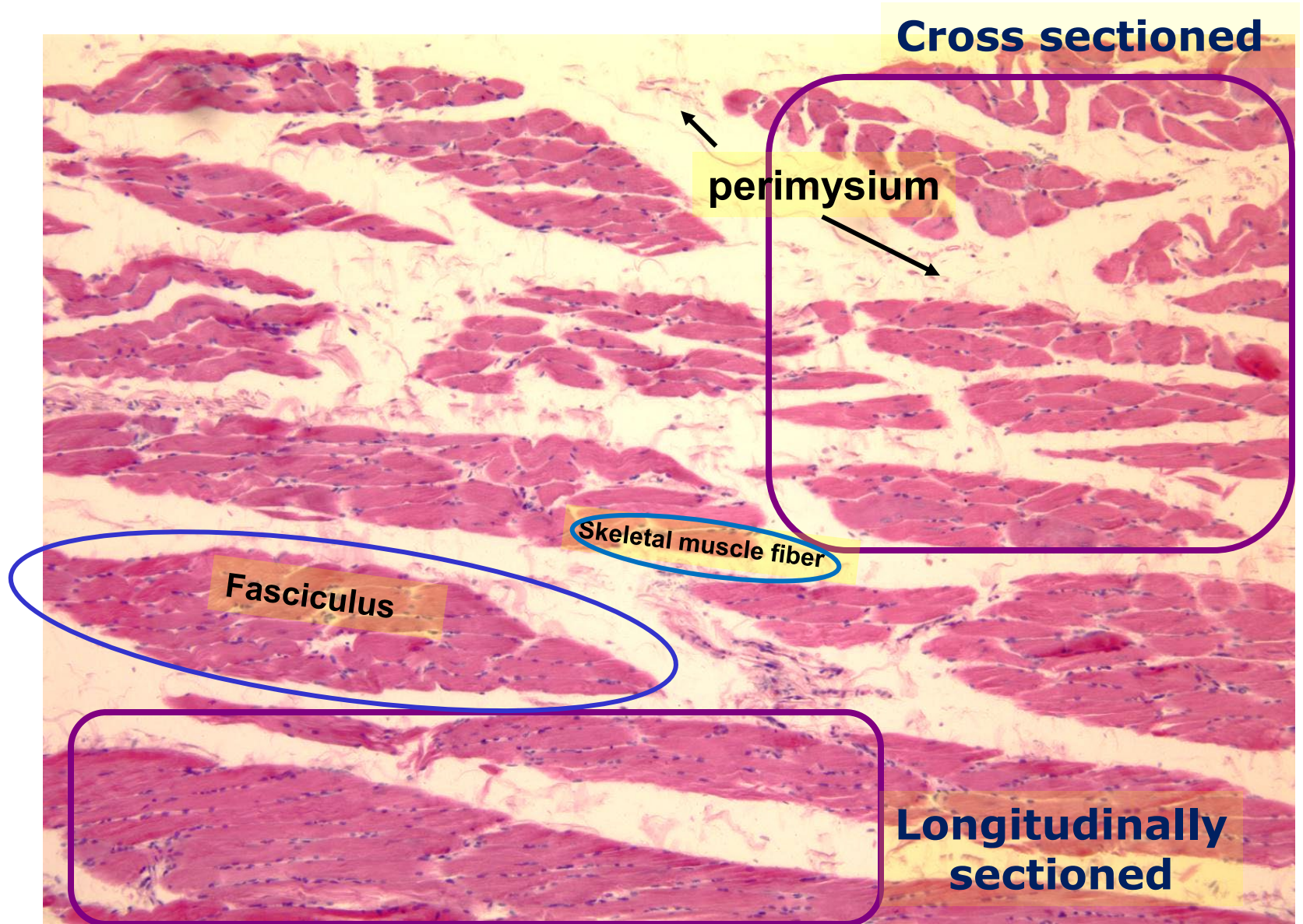


**#6** Skeletal muscle – striation/voluntary, HE, human



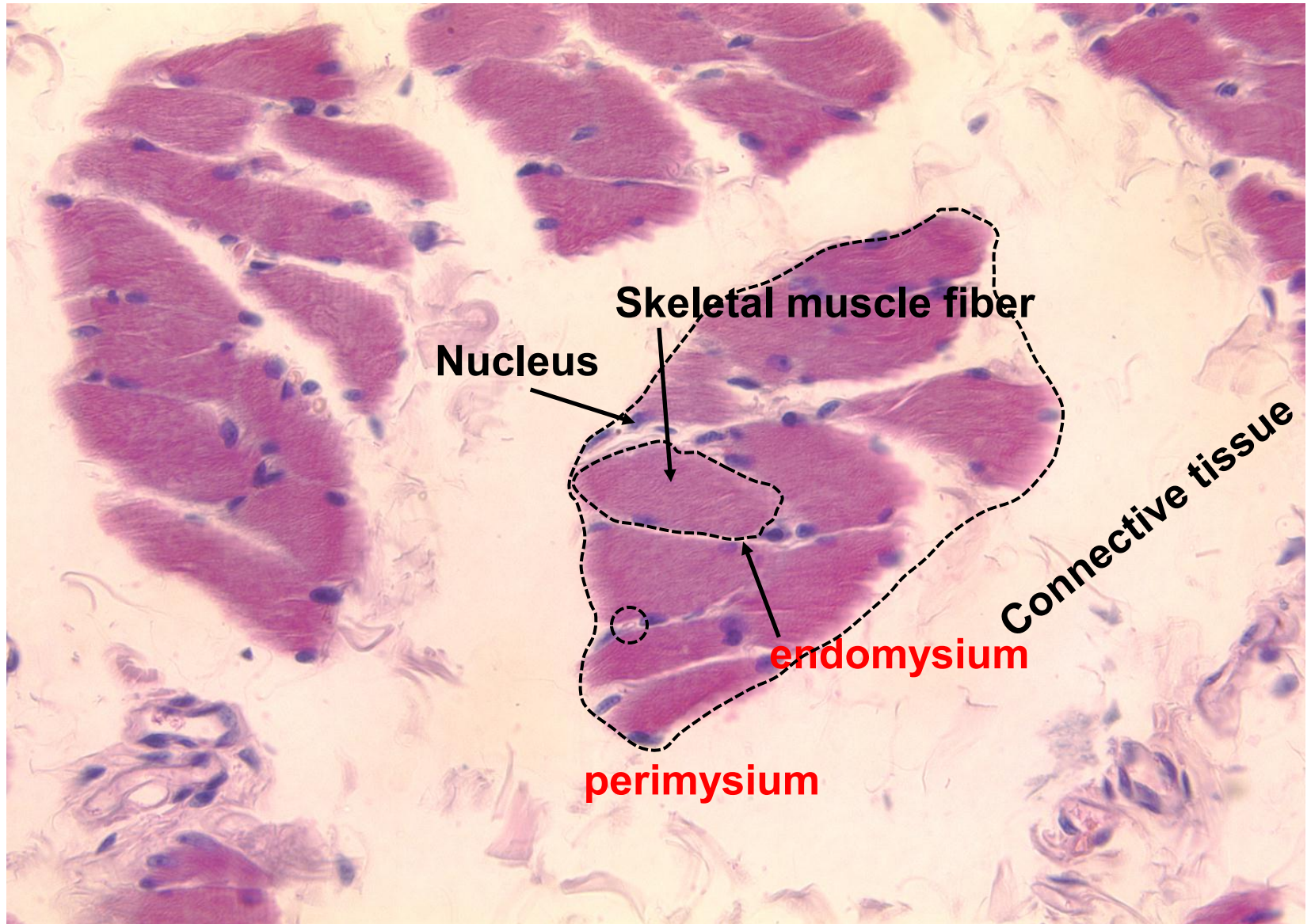
**x 40**

**#6** Skeletal muscle – striation/voluntary, HE, human



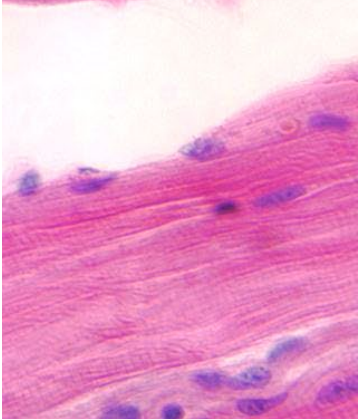
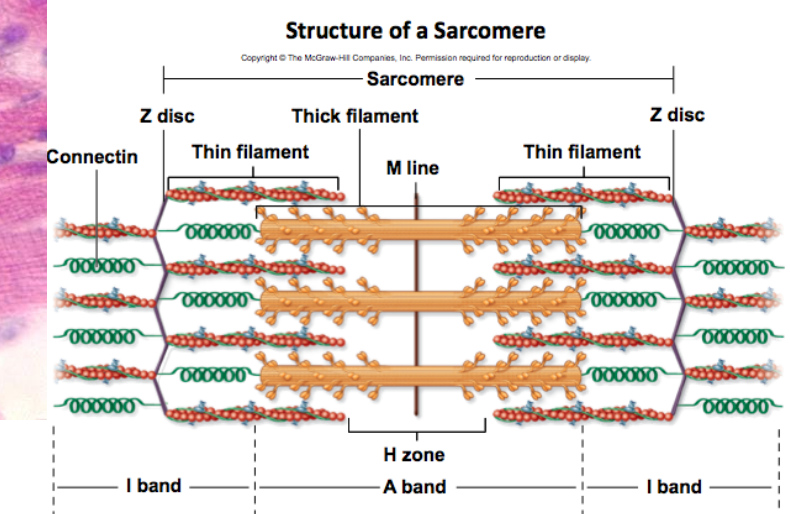
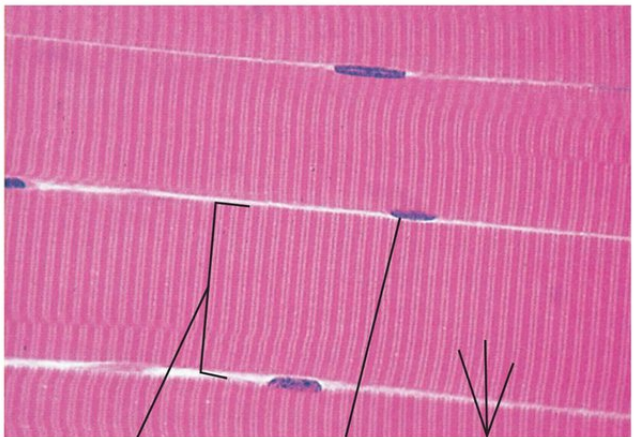
x100

**#6 Skeletal muscle – striation/voluntary, HE, human , Cross**



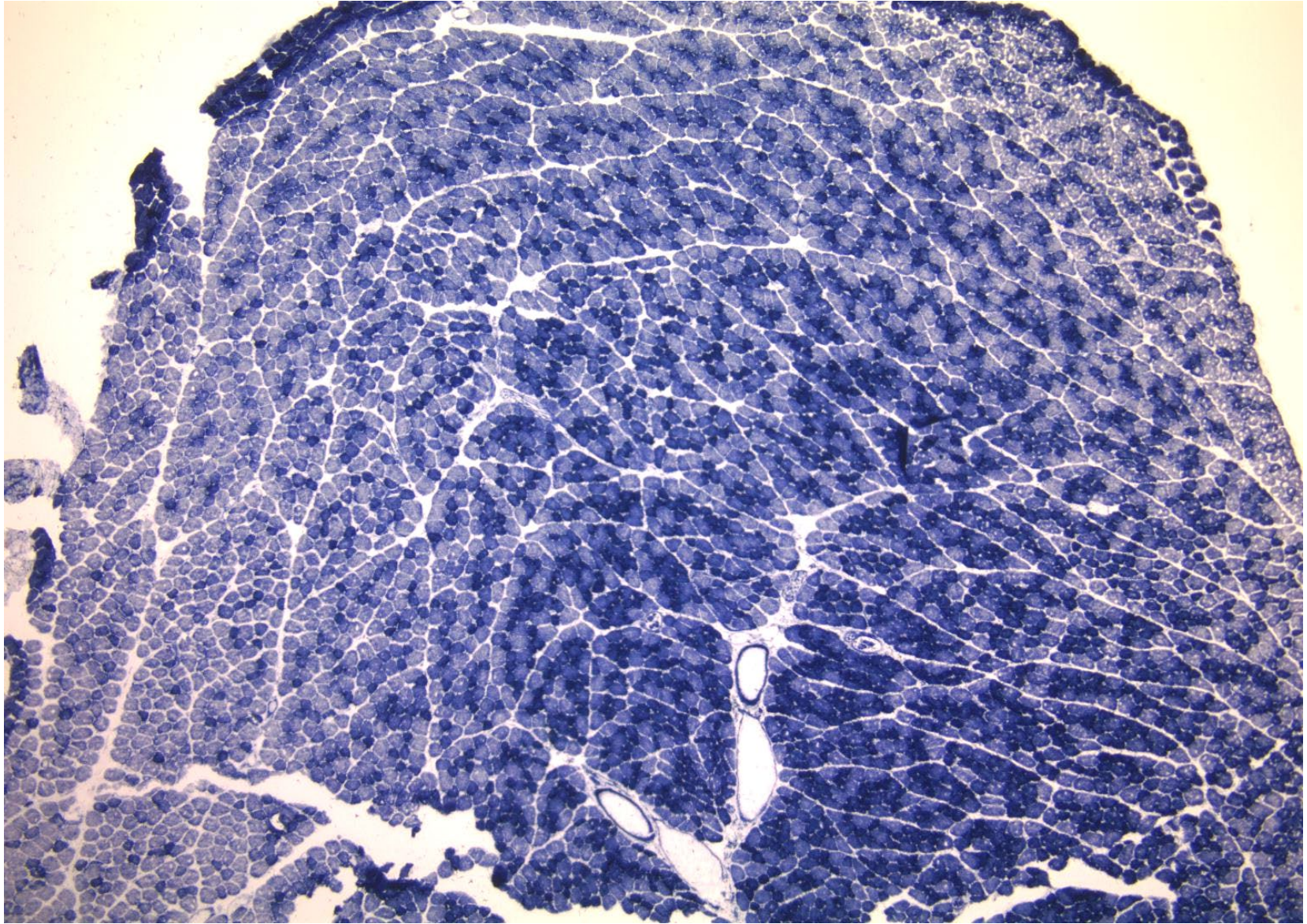
**x 400**

#6 Skeletal muscle – striation / voluntary, HE, human , longitudinally



x 400

**#8** Skeletal muscle – striation / voluntary, NADH-TR, rat

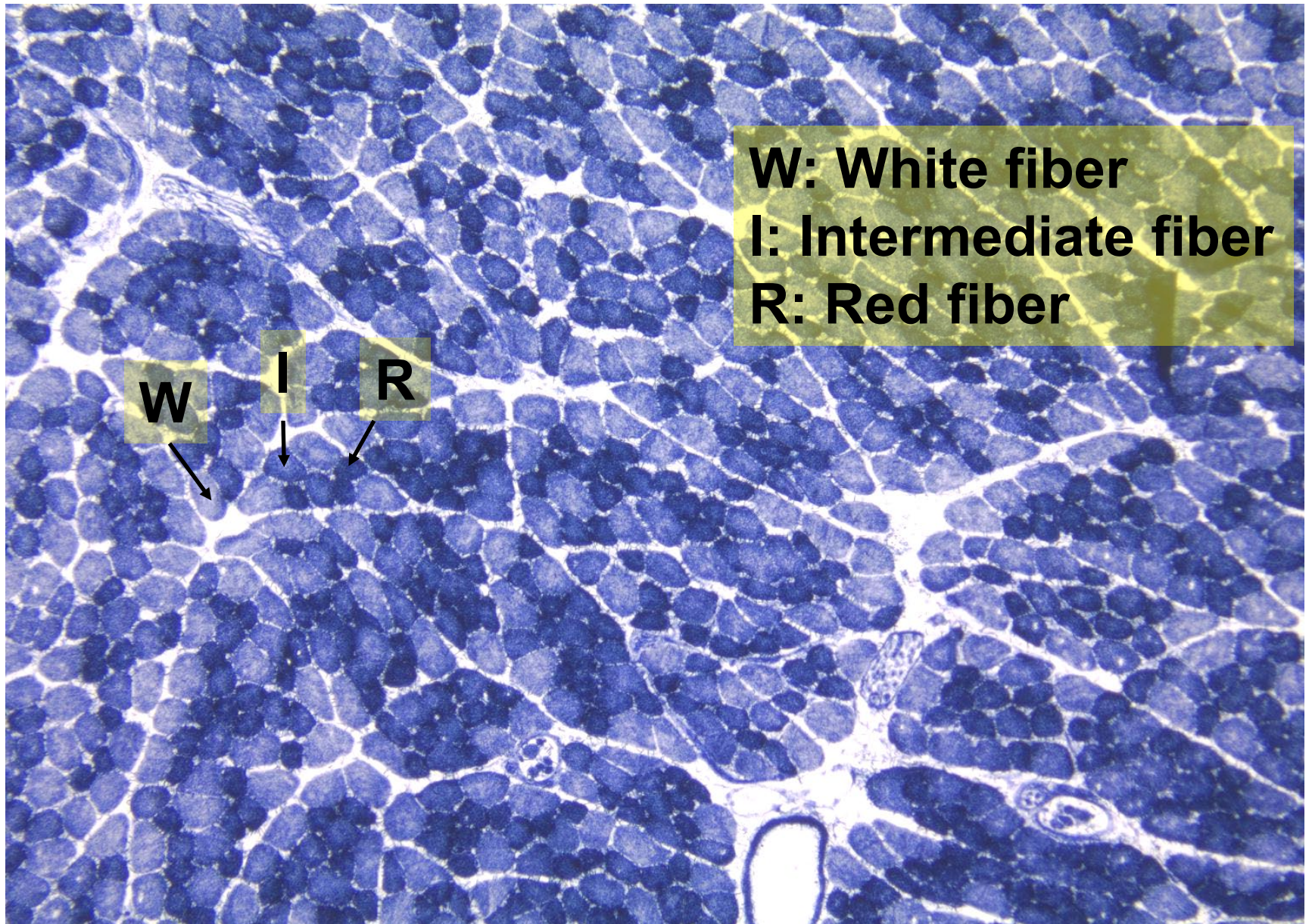


**NADH-TR staining** : stains enzyme inside mitochondria  
(Nicotinamide adenine dinucleotide-tetrazolium reductase) **x40**

# Classification of skeletal muscle

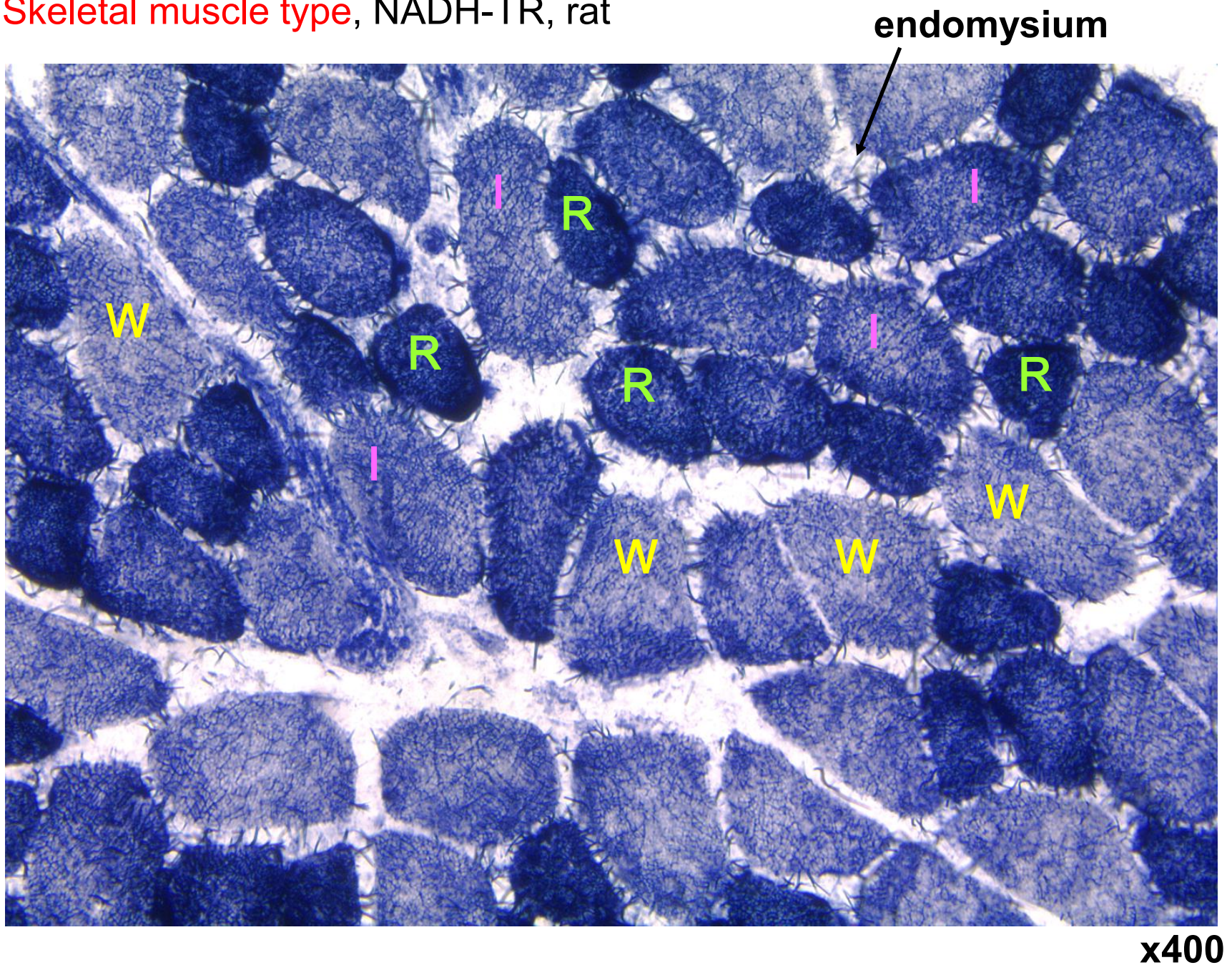
|              | Skeletal Muscle                     |                      |                                     |
|--------------|-------------------------------------|----------------------|-------------------------------------|
|              | 백색근                                 | 중간근                  | 적색근                                 |
| 직경           | 굵음                                  | 중간                   | 가늘                                  |
| mitochondria | 적음<br>(cytochrome 적음,<br>ATPase 적음) | 풍부                   | 풍부<br>(cytochrome 풍부,<br>ATPase 풍부) |
| myoglobin 수  | 적음                                  | 풍부                   | 풍부                                  |
| 수축속도         | Fast                                | Fast                 | Slow                                |
| 에너지대사        | Glycolytic                          | Oxidative-glycolytic | Oxidative phosphorylation           |
| 운동능력         | 순발적인 운동                             | 느리고 지구력<br>요구되는 운동   | 느리고 지구력<br>요구되는 운동                  |
| NADH-TR 염색   | 제일 연합                               | 중간 진함                | 제일 진함                               |

#8 3 types of skeletal muscle fiber, NADH-TR, rat



x100

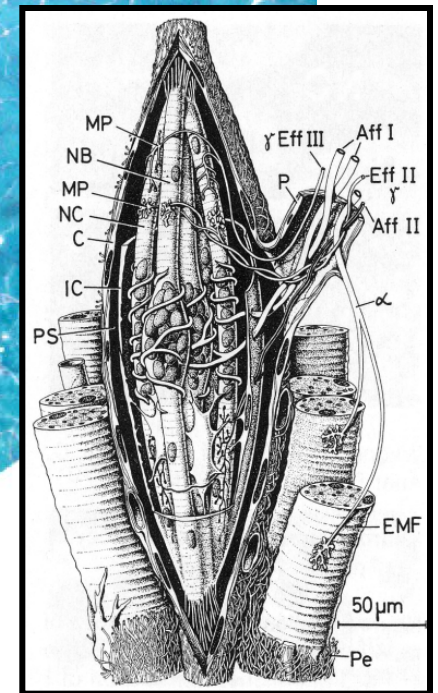
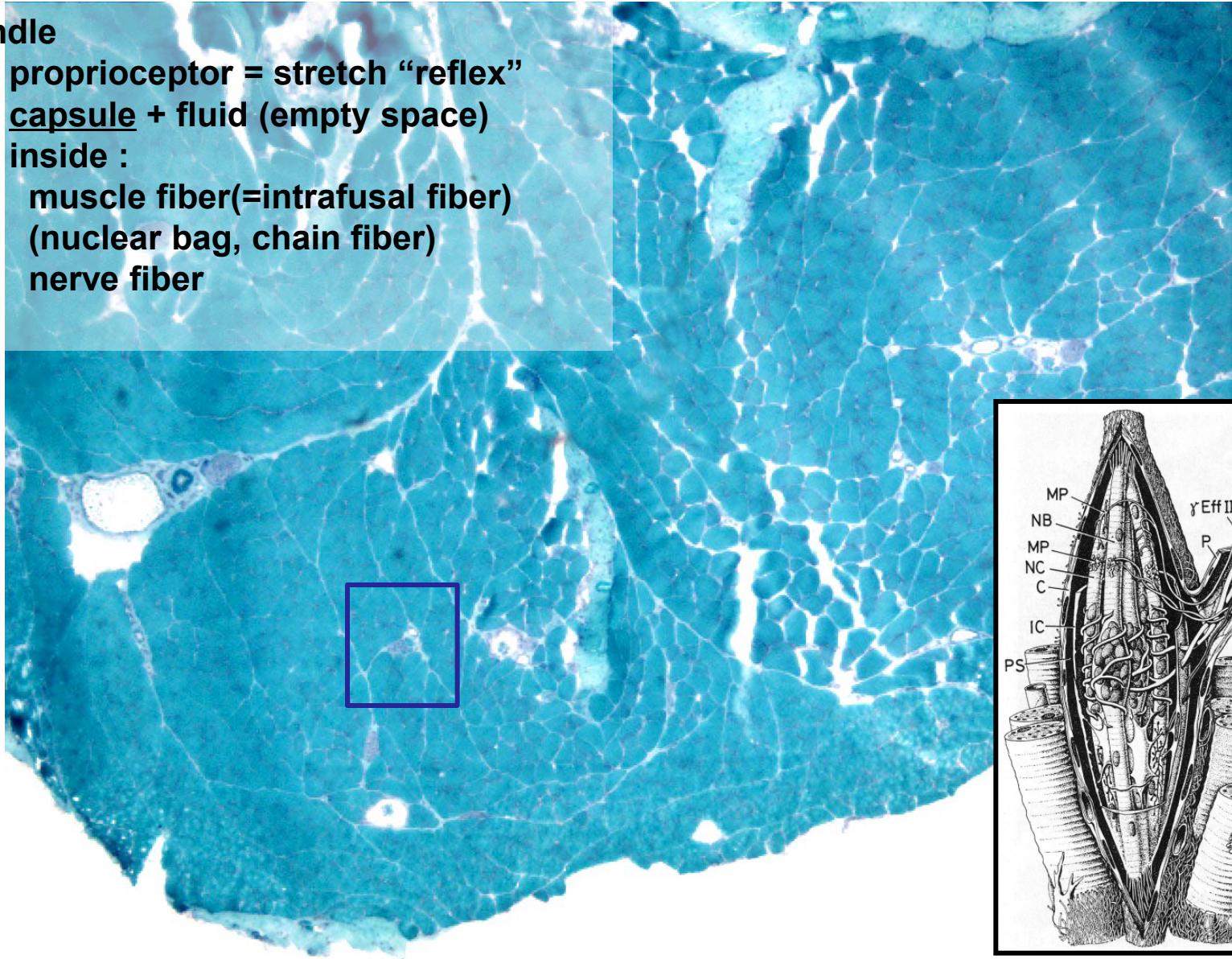
#8 Skeletal muscle type, NADH-TR, rat



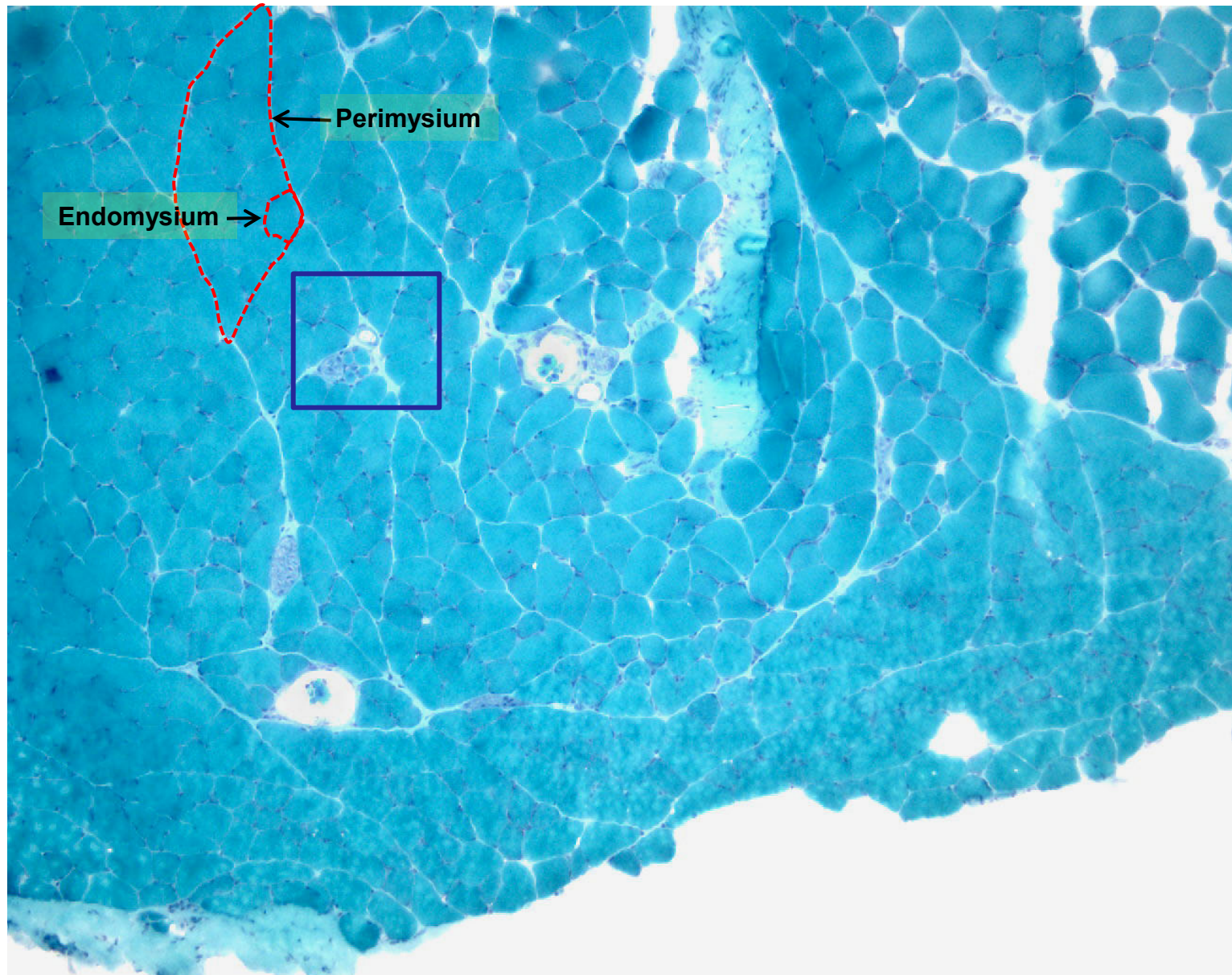
# #7 Skeletal muscle, modified Gomori trichrome, rat - Neuromuscular spindle (신경근육방추)

## NM spindle

proprioceptor = stretch “reflex”  
capsule + fluid (empty space)  
inside :  
muscle fiber(=intrafusal fiber)  
(nuclear bag, chain fiber)  
nerve fiber

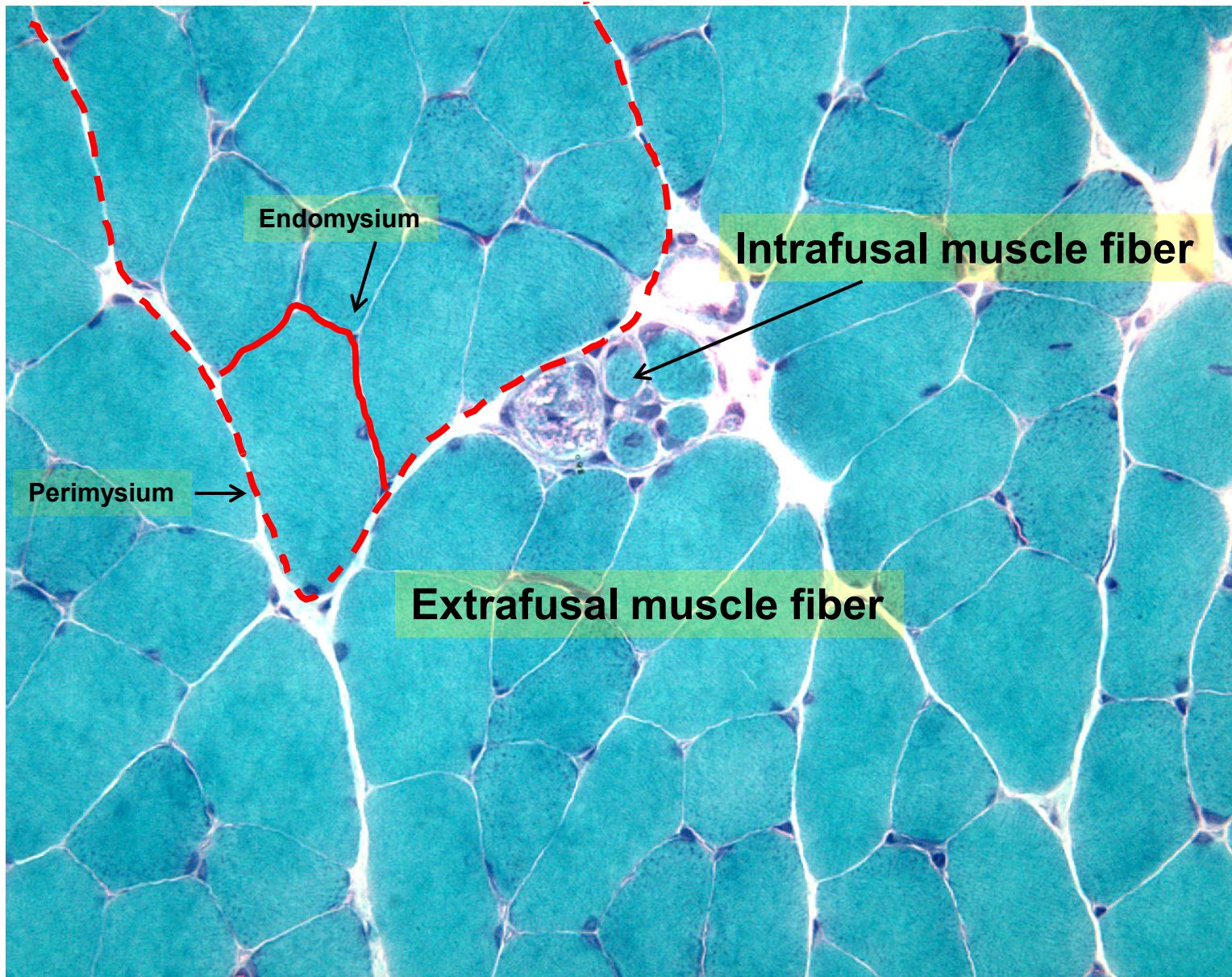


# #7 Skeletal muscle, modified Gomori trichrome, rat - Neuromuscular spindle



×100

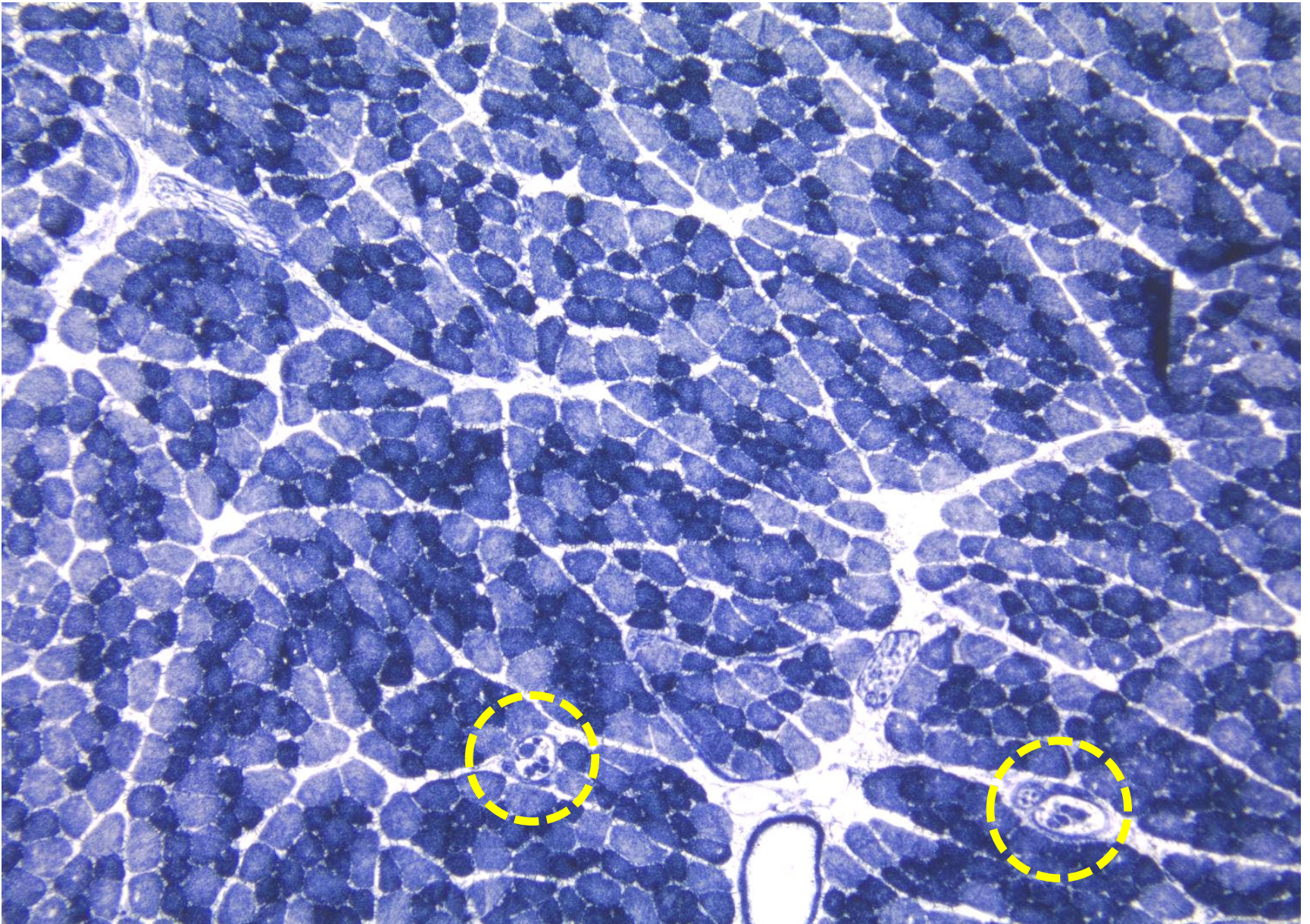
#7 Skeletal muscle, modified Gomori trichrome, rat- Neuromuscular spindle ★



x400

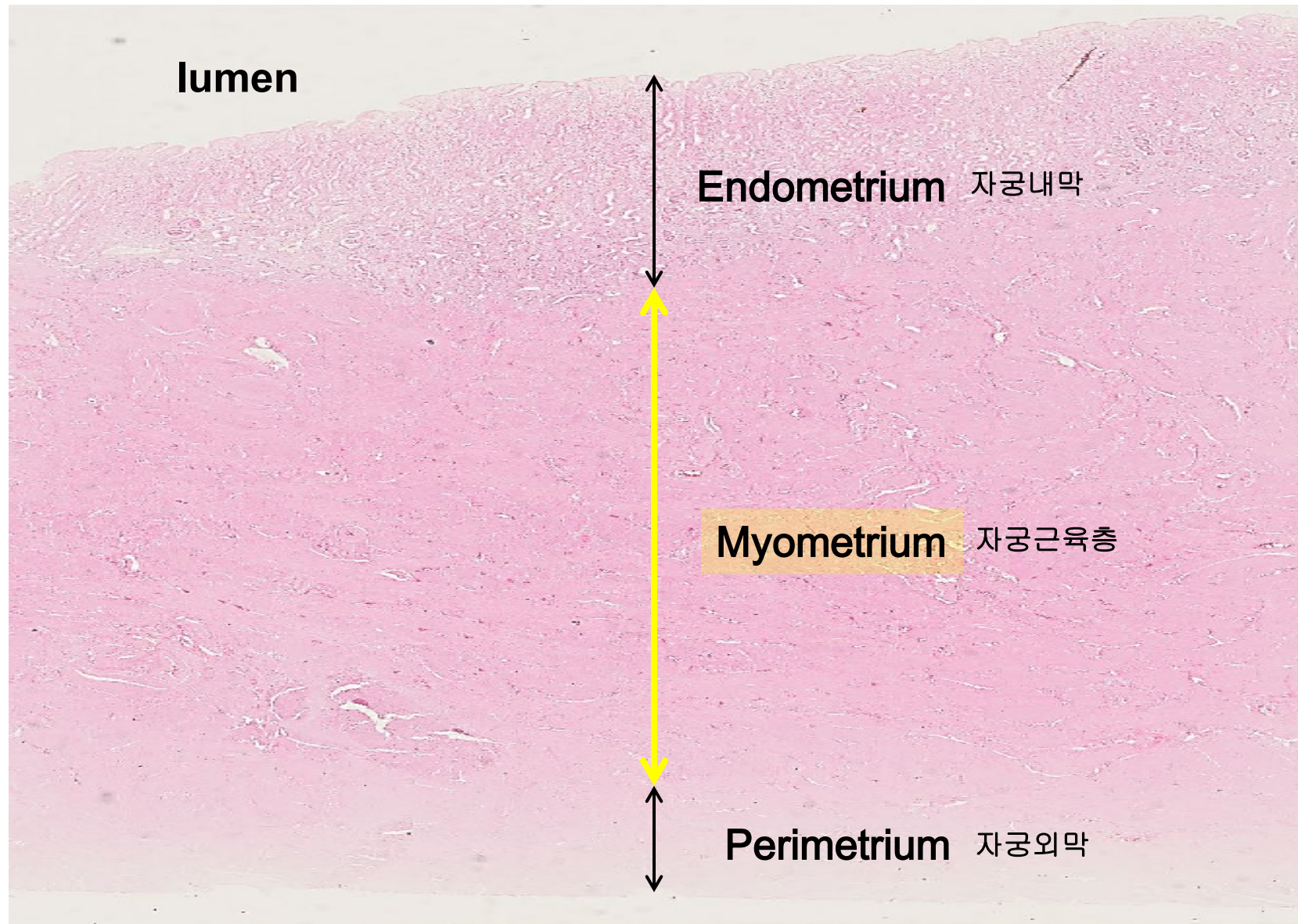
#8 NADH-TR, rat

Neuromuscular spindle



x100

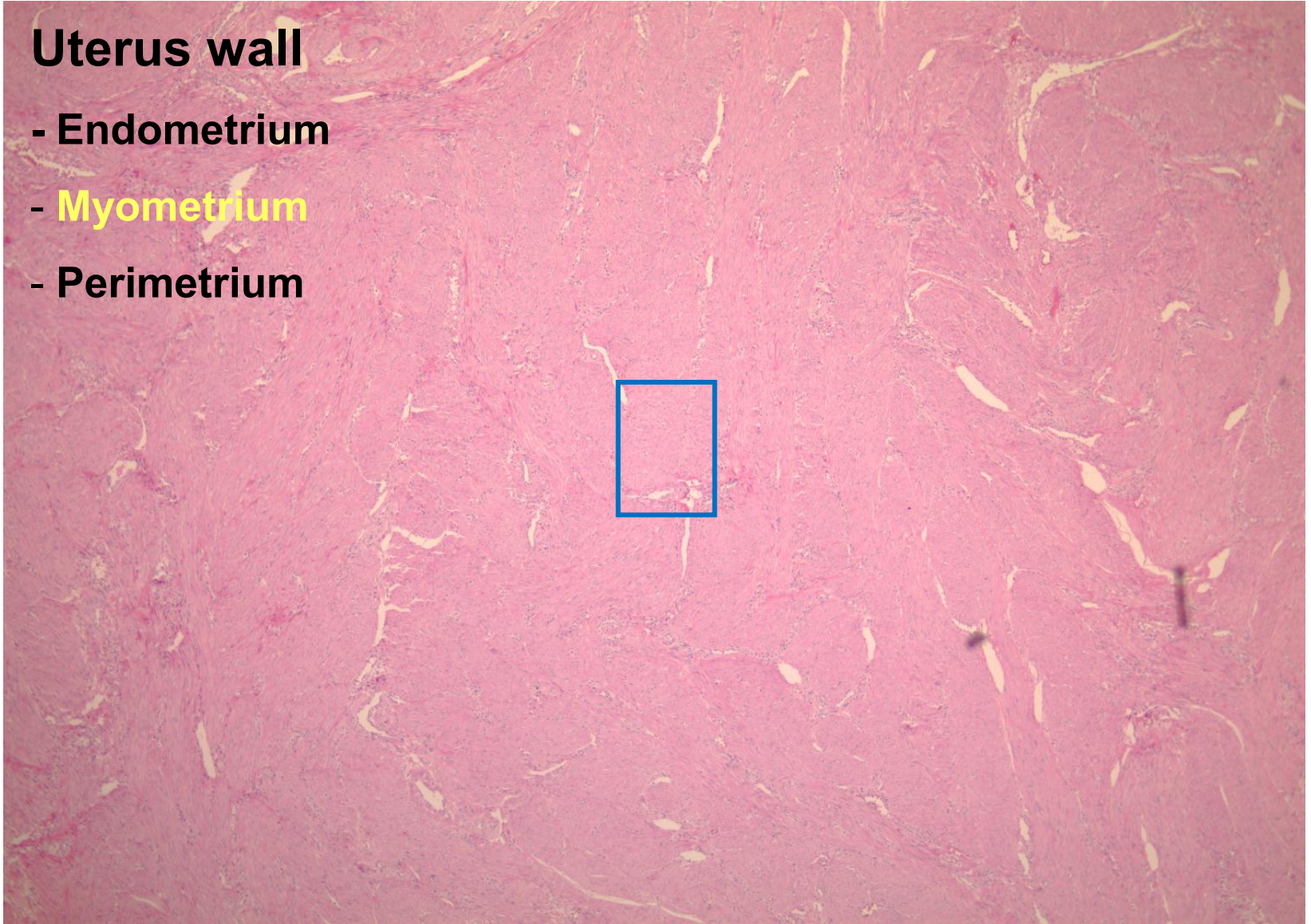
**#57 Uterus, monkey, HE, Smooth muscle – no striation / involuntary**



**#58** Uterus, HE, human, **Smooth muscle – no striation / involuntary**

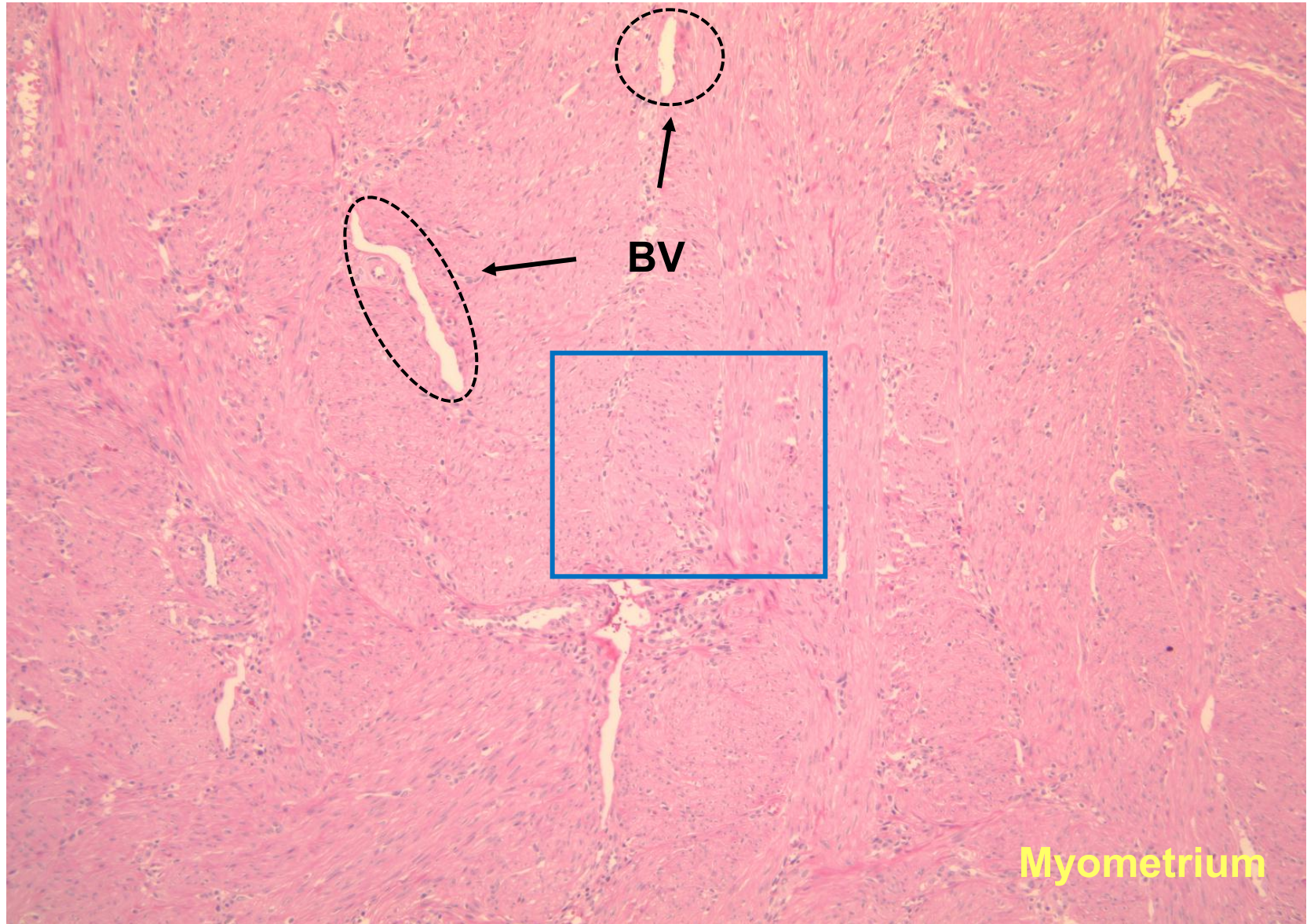
**Uterus wall**

- Endometrium
- **Myometrium**
- Perimetrium

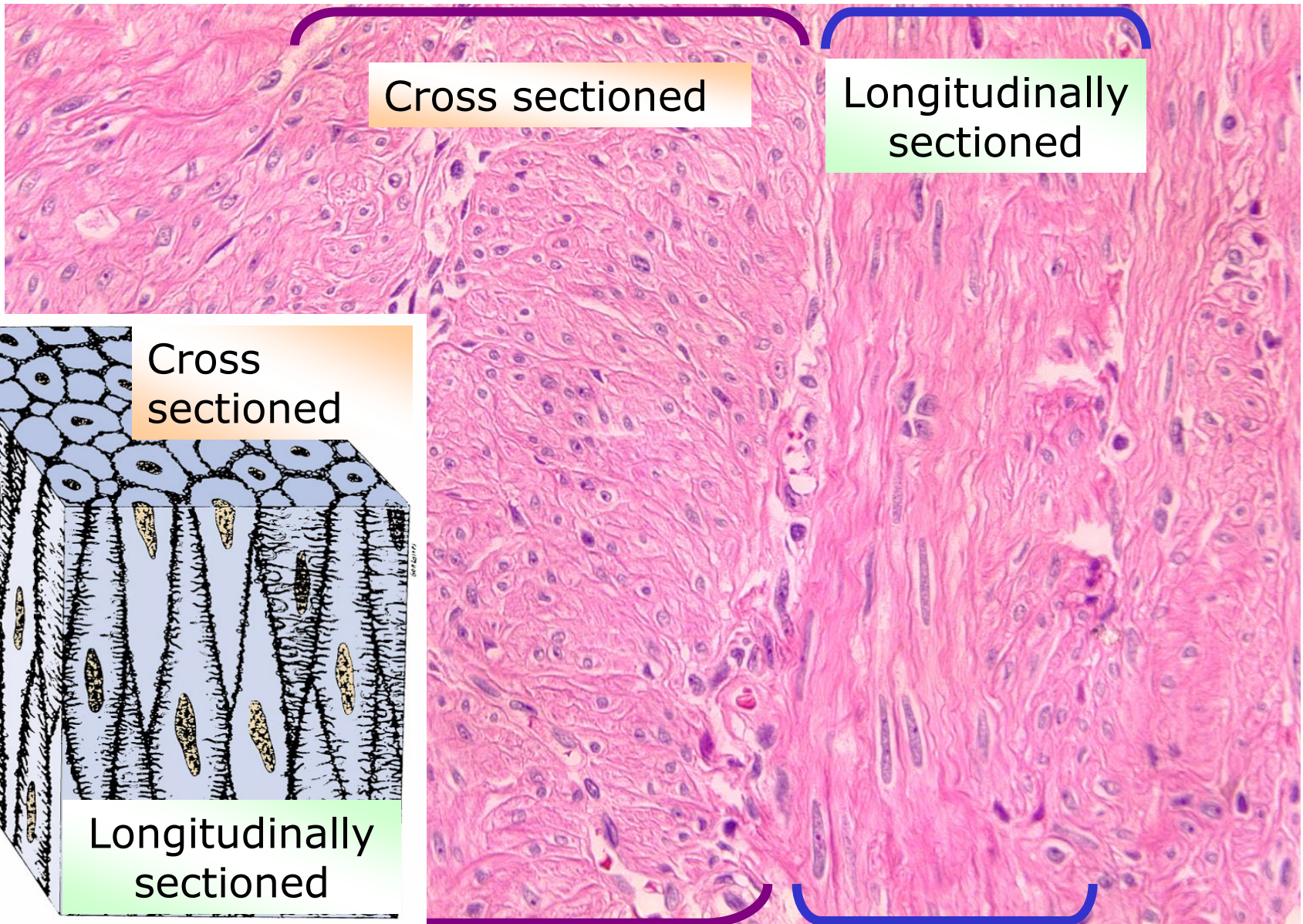


**x40**

**#58** Uterus, HE, human, **Smooth muscle -no striation / involuntary**

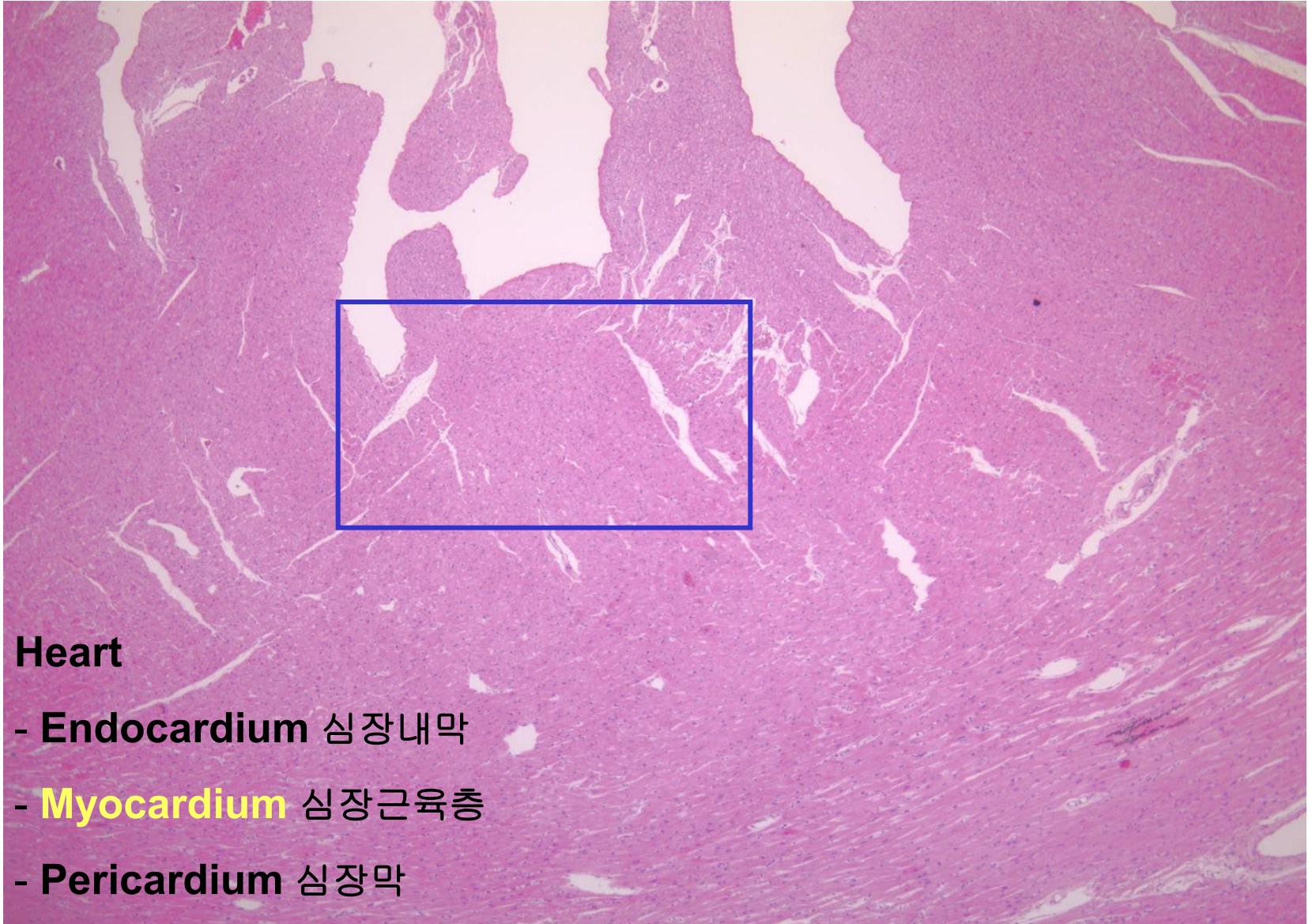


**x100**



x400

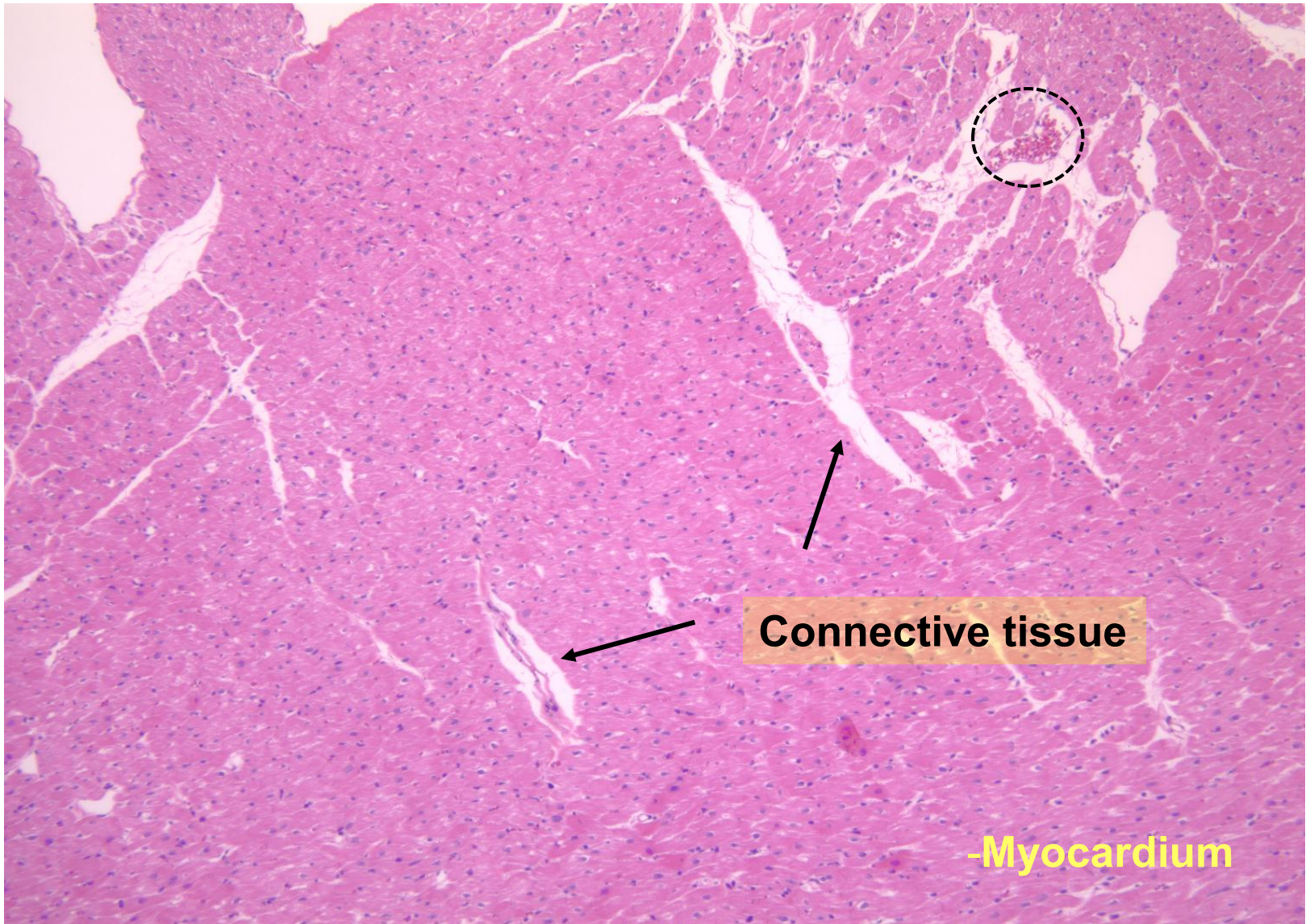
#73 Heart, HE, monkey - Cardiac muscle -striation / involuntary, longitudinal



**Heart**

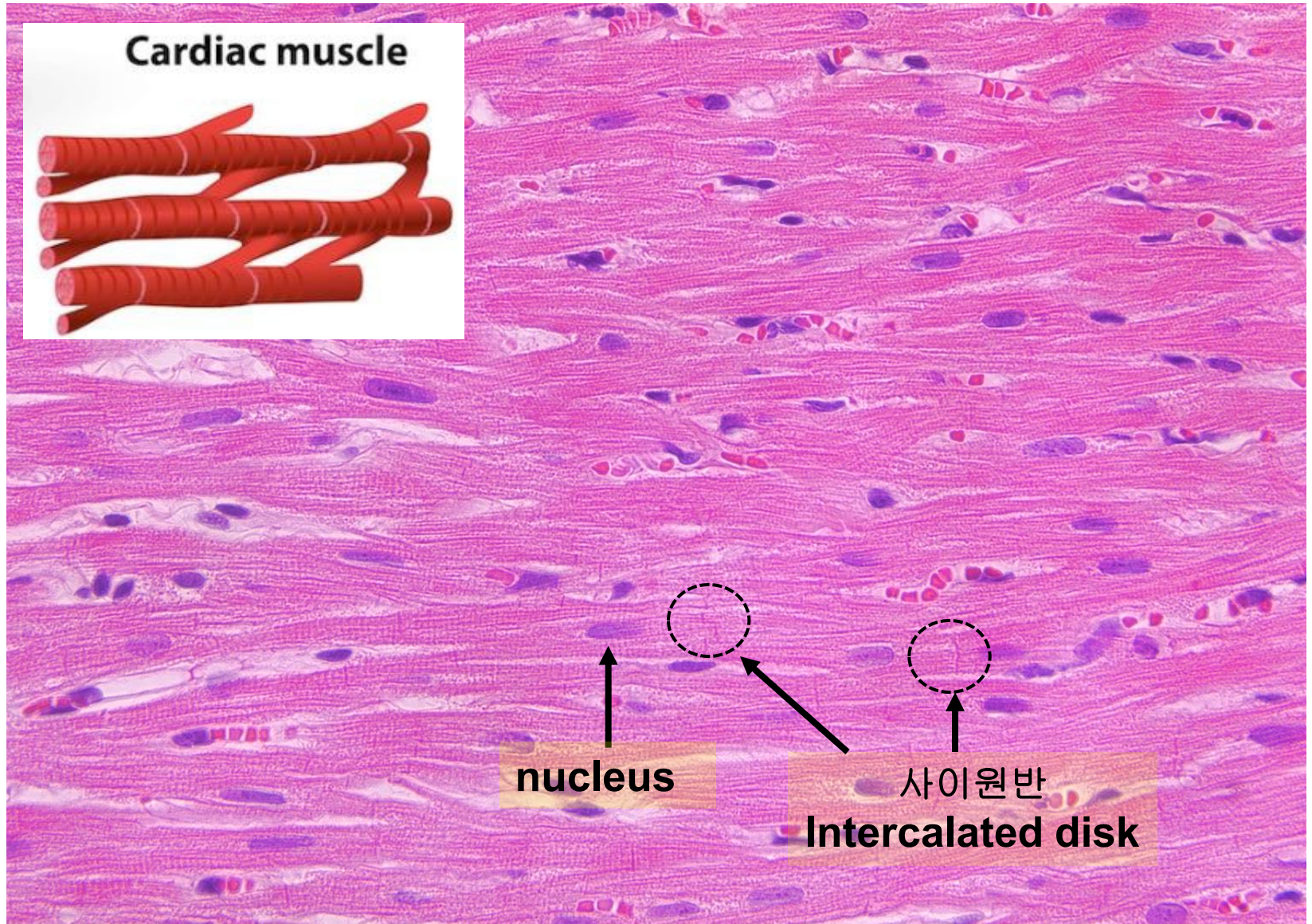
- Endocardium 심장내막
- **Myocardium** 심장근육층
- Pericardium 심장막

**#73** Heart, HE, monkey - Cardiac muscle -striation / involuntary, longitudinal

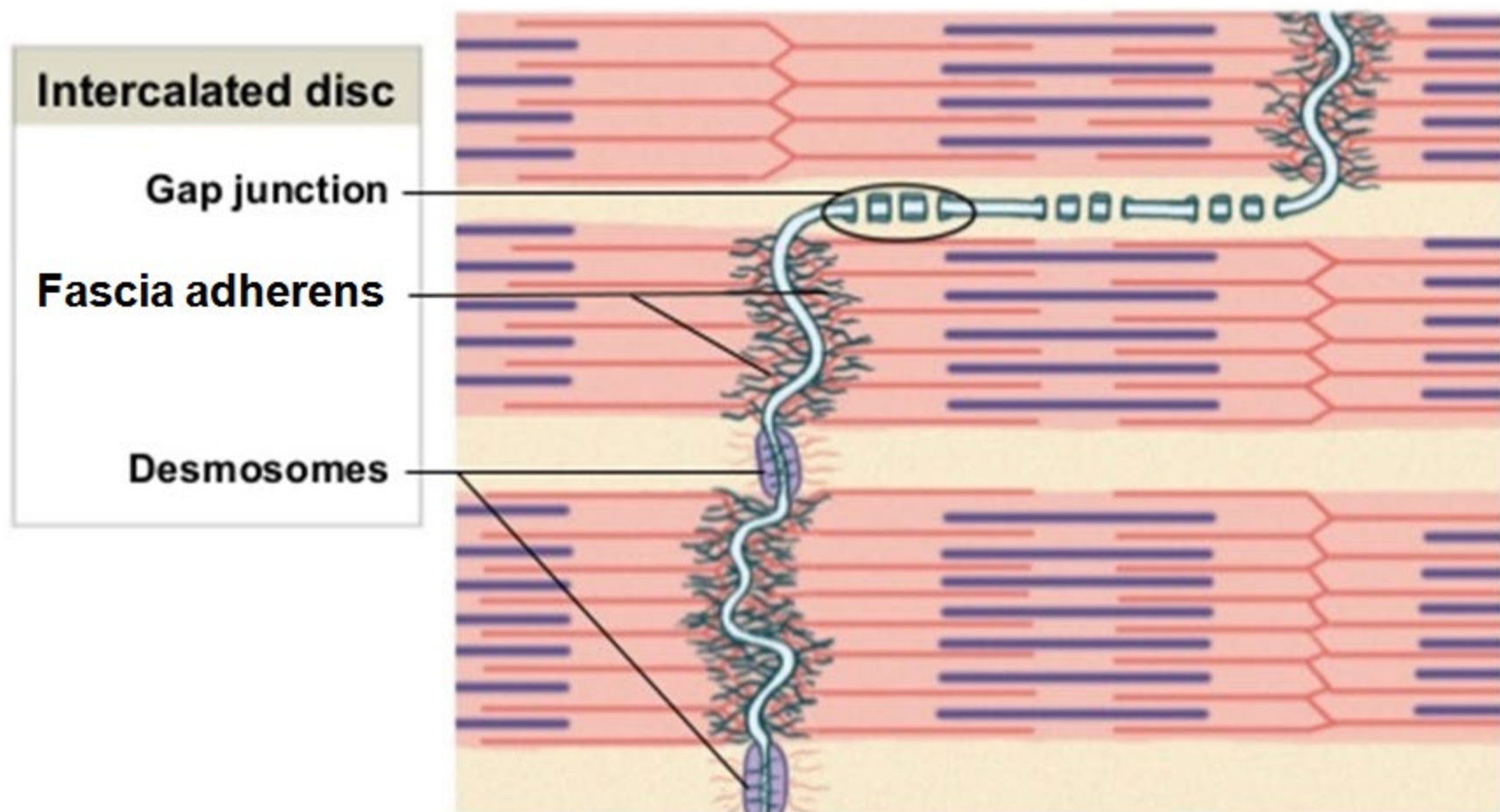


**x100**

#73 Heart, HE, monkey - Cardiac muscle -striation / involuntary, longitudinal ★

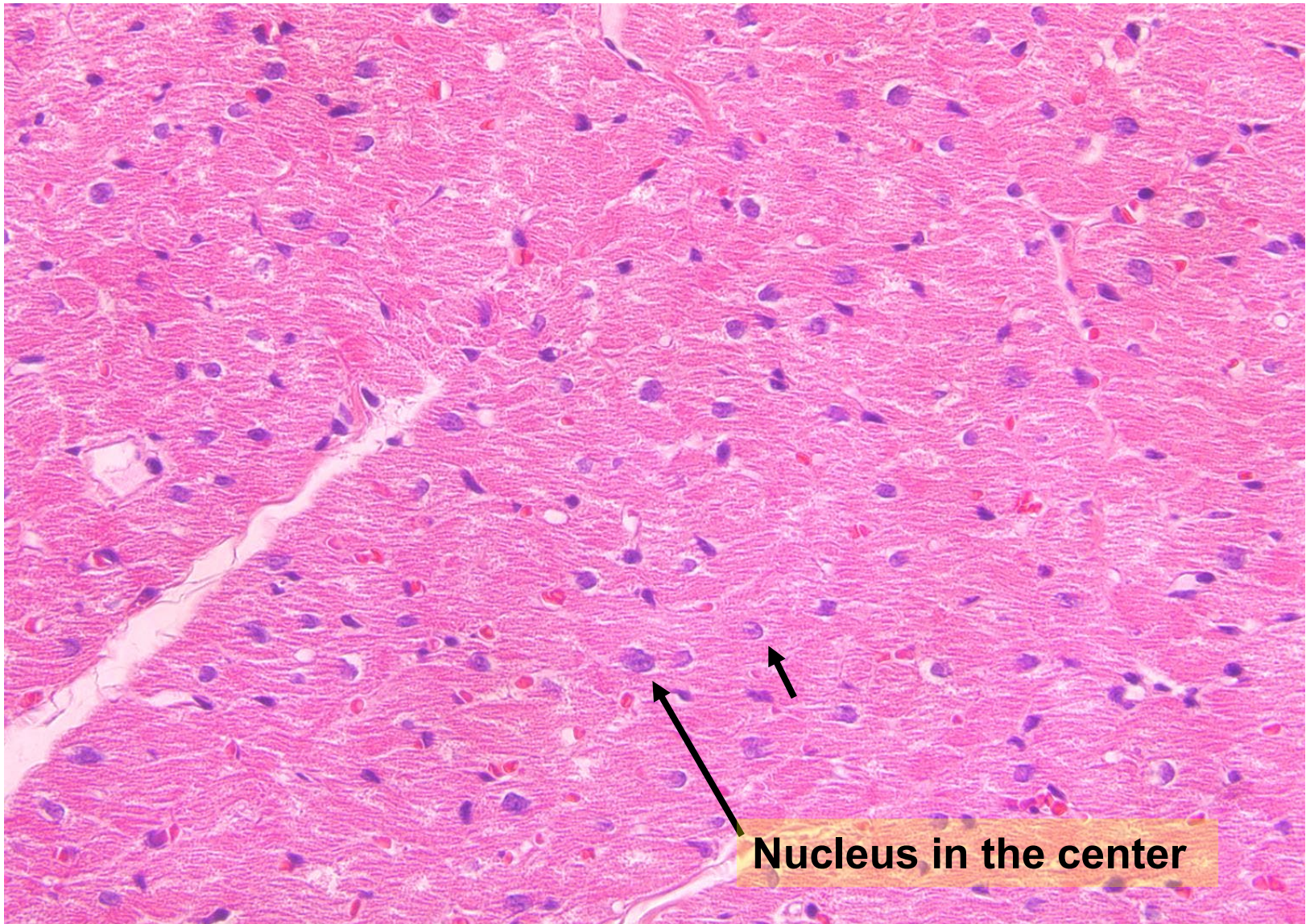


x400



**b** Structure of an intercalated disc

**#73** Heart, HE, monkey -Cardiac muscle -striation / involuntary, cross



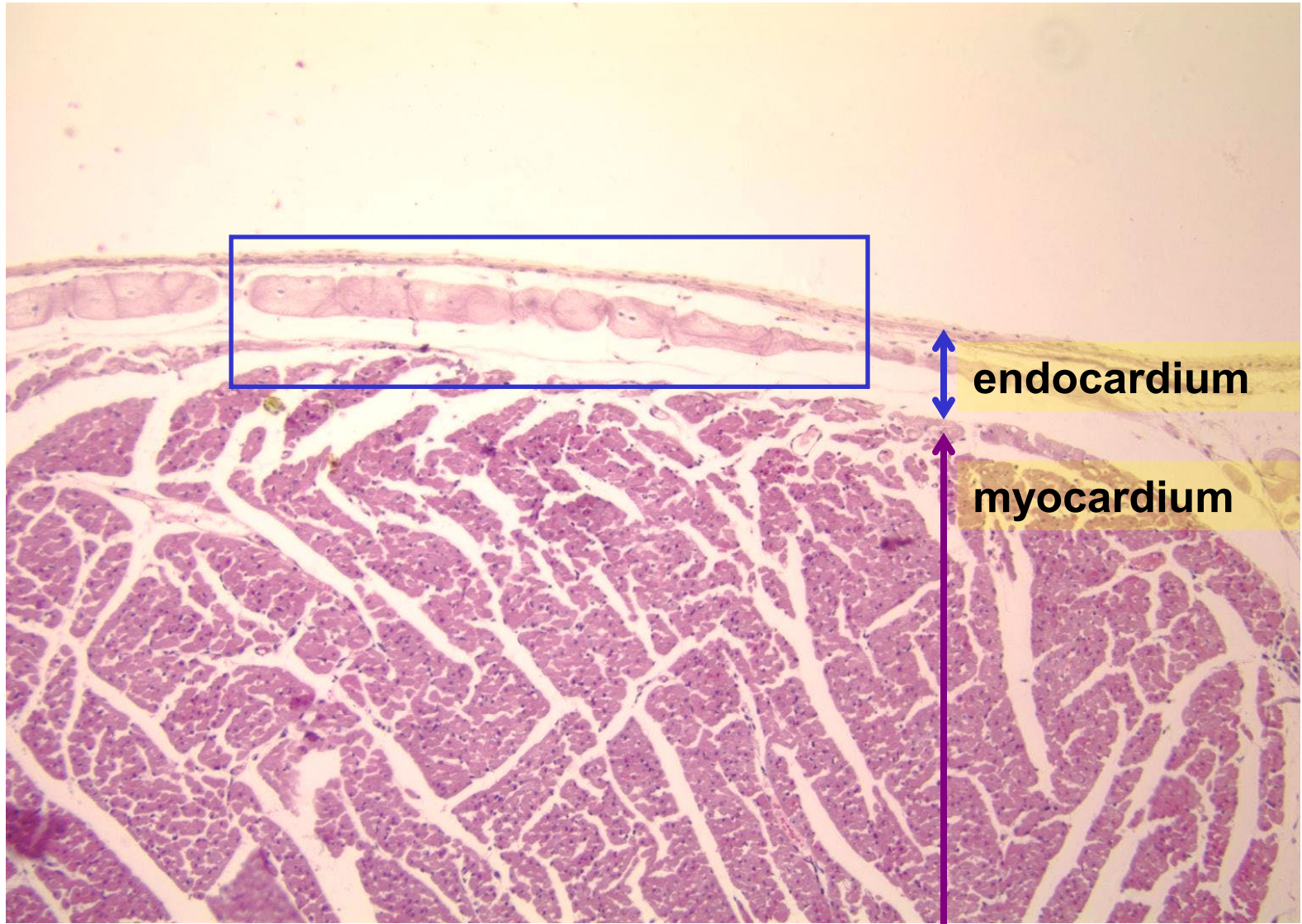
**x400**

# #19 Heart, HE, dog, Purkinje Fiber

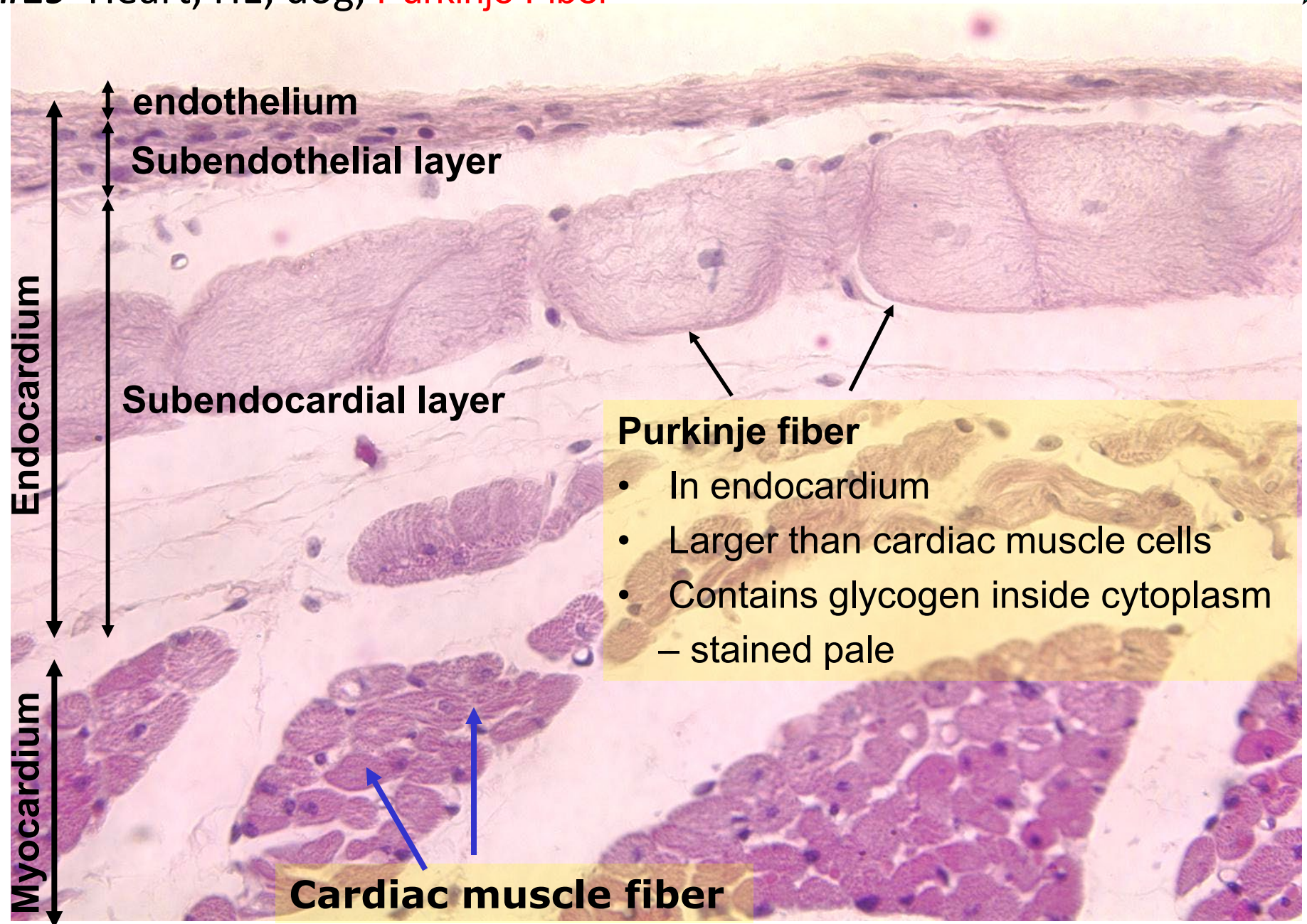


x40

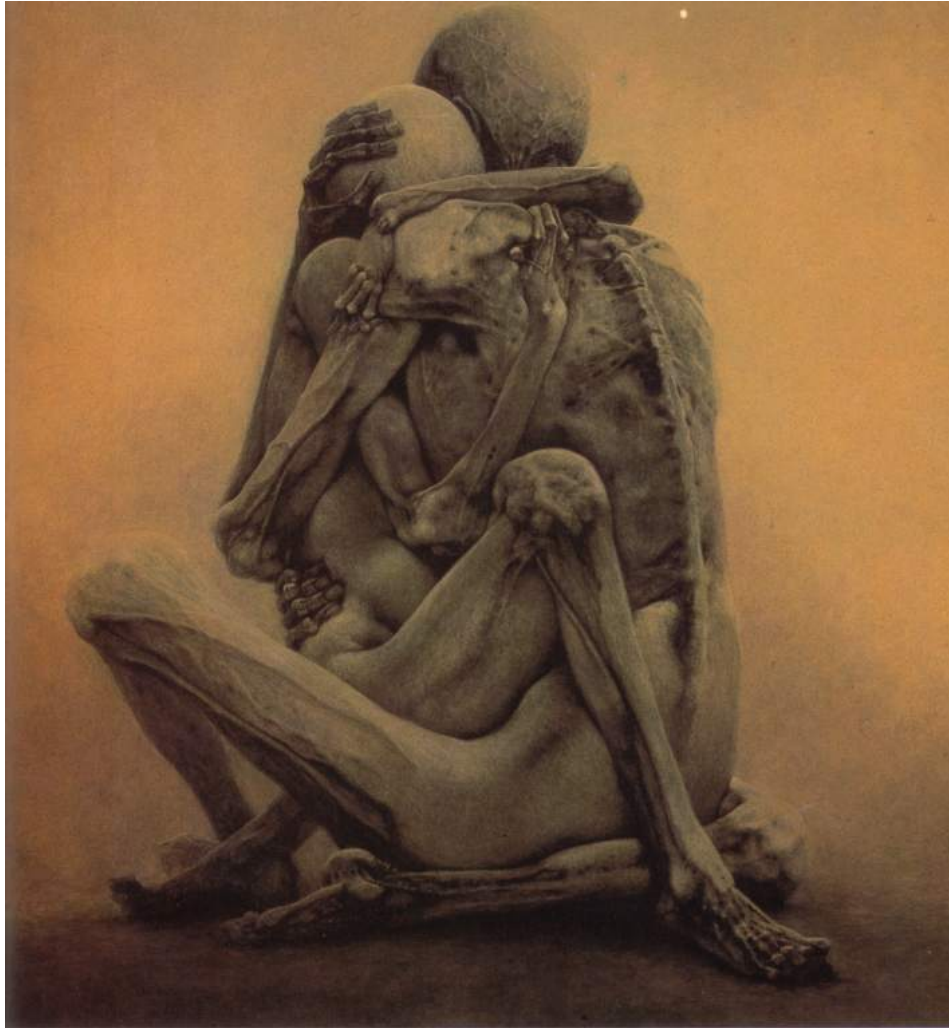
# #19 Heart, HE, dog, Purkinje Fiber



x100



# Cartilage and Bone



**#28 Trachea, HE, monkey, ×400 - Hyaline cartilage**

**#76 Larynx (epiglottis), HE, human - Elastic cartilage**

(#79 Epiglottis, human, Orcein)

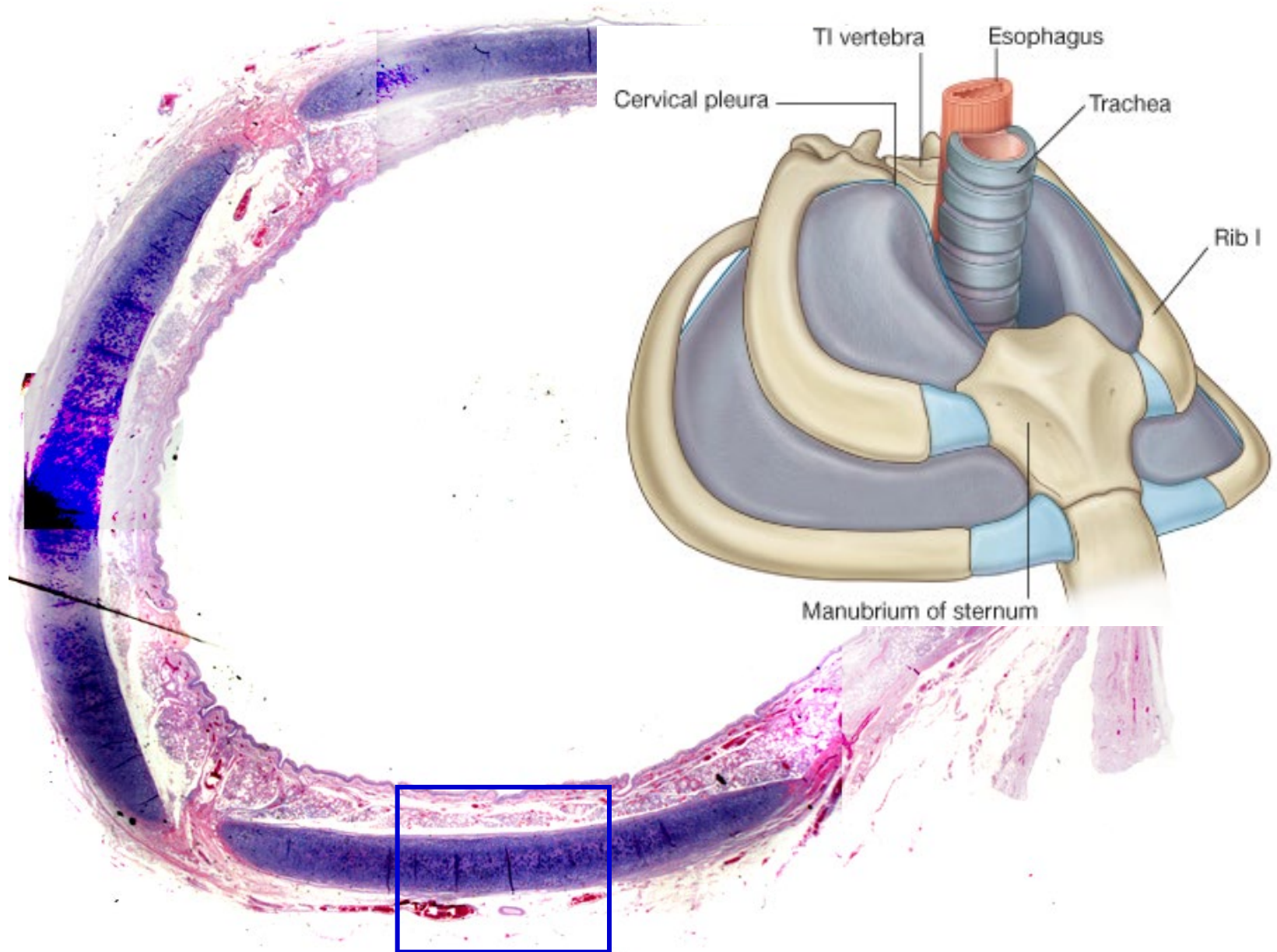
**#5 Intervertebral disc, HE, human, ×400 - Fibrocartilage**

# Cartilage (연골)

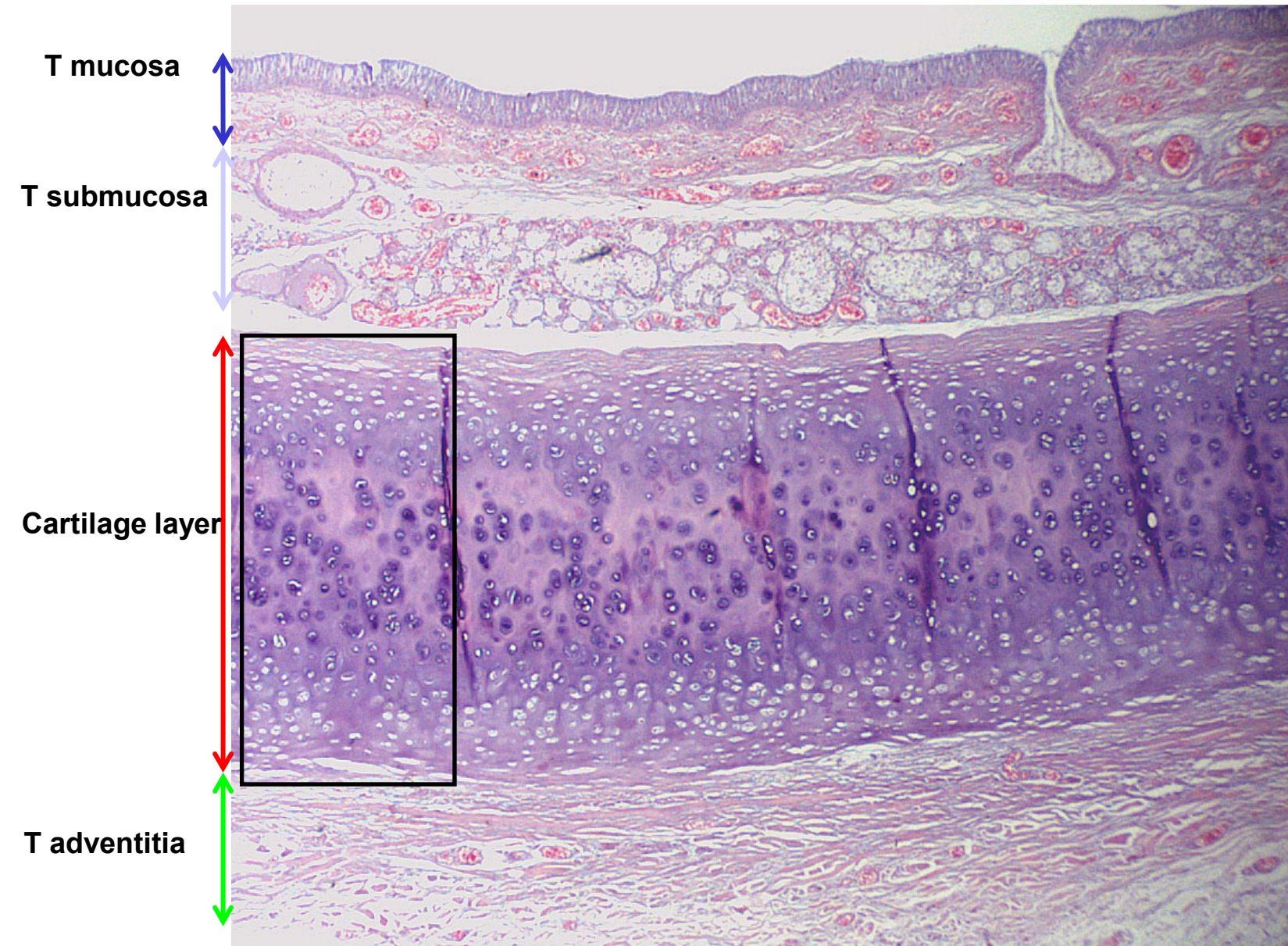
- 특수결합조직
- 종류
  - 1) 유리연골(hyaline cartilage) : II형 collagen
  - 2) 탄력연골(elastic cartilage) : II형 collagen + elastic fiber(탄력섬유)
  - 3) 섬유연골(fibrocartilage) : I형 collagen
- 혈관, 신경, 림프관이 없다.
- 연골막(perichondrium) - 섬유연골 제외  
; 치밀결합조직, 연골에 영양분 공급

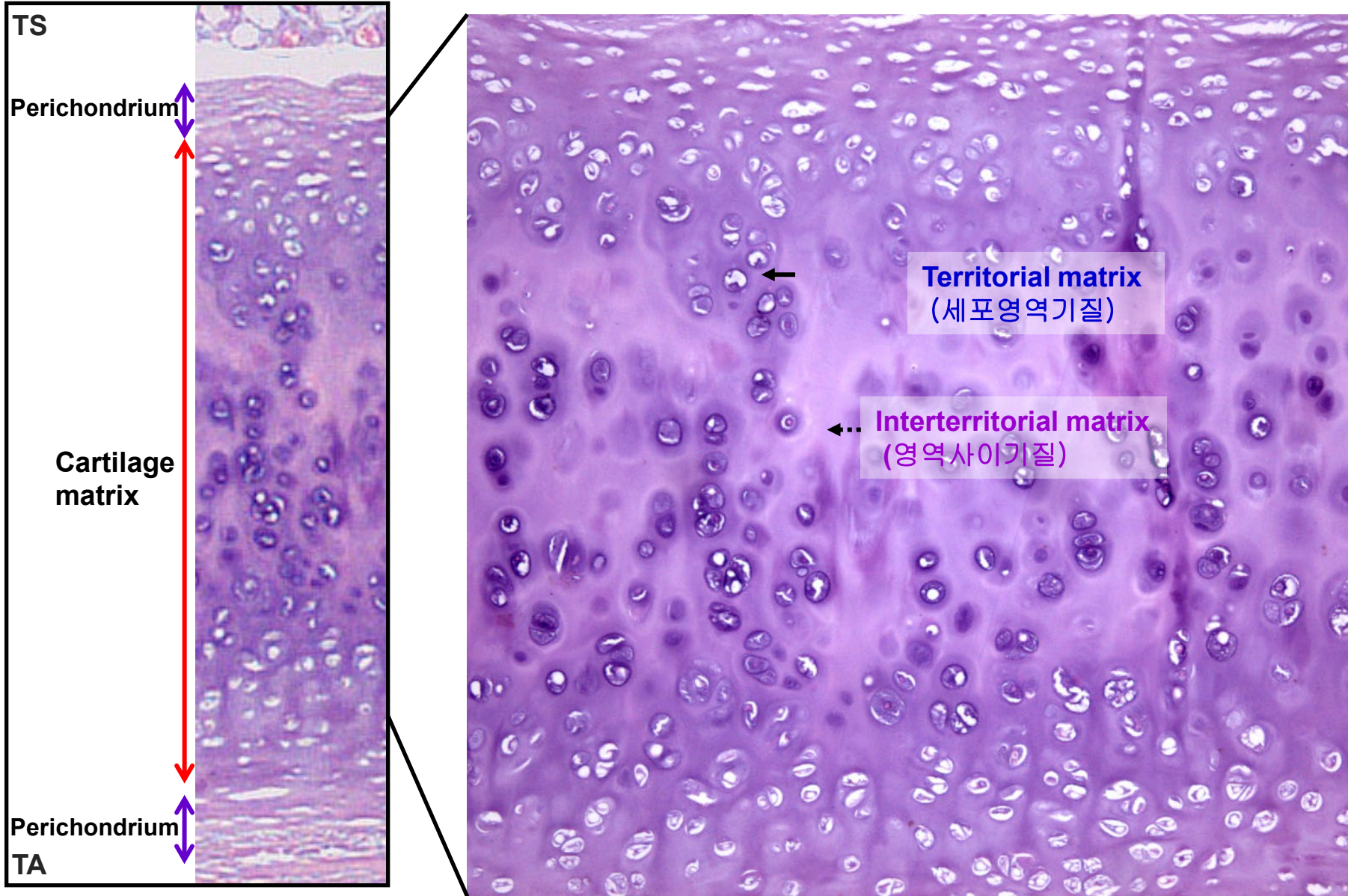
# #28 Trachea, monkey, HE,

## Hyaline cartilage (유리연골)



**Stereomicroscopic view**







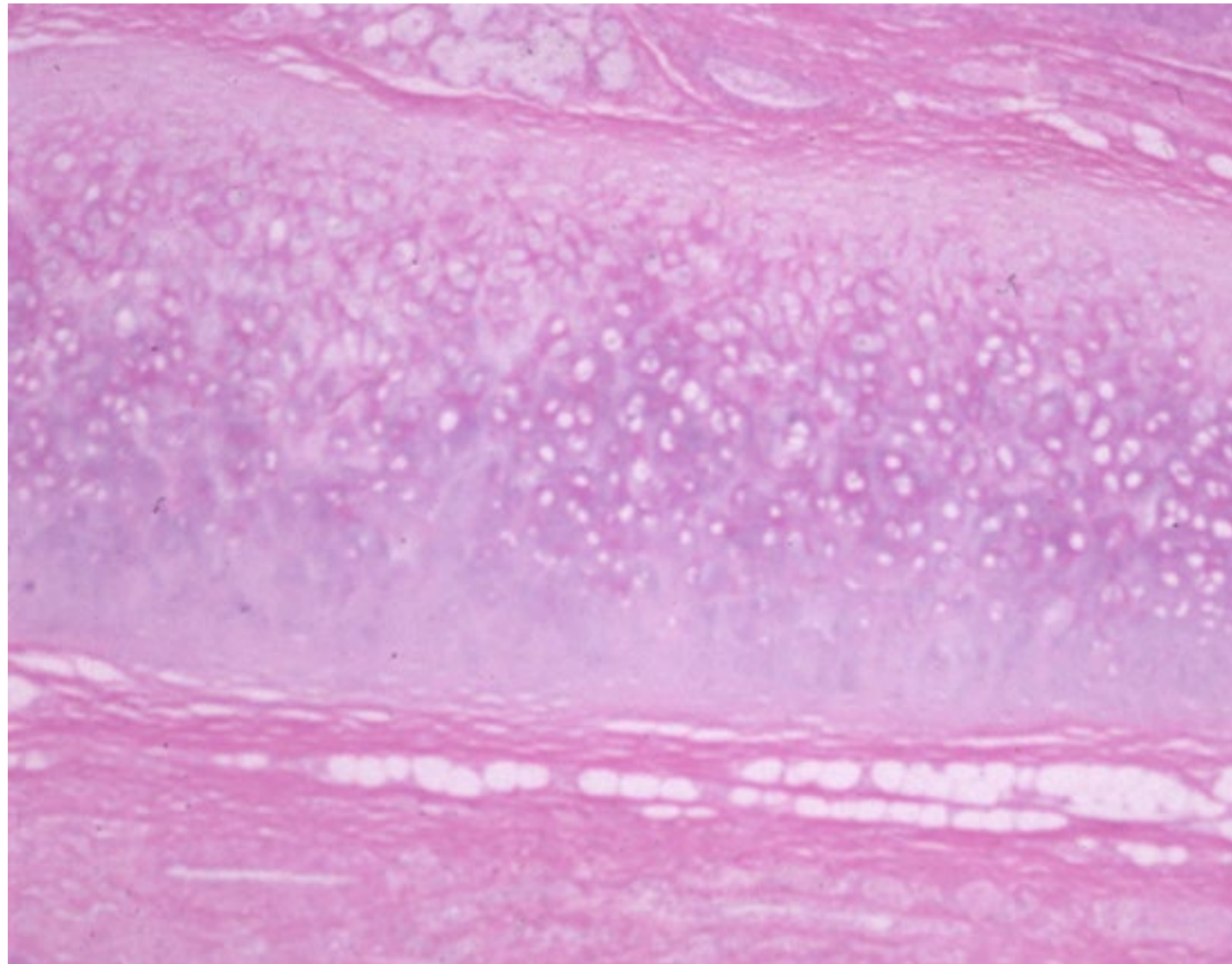
**Territorial matrix**  
(세포영역기질)

**Isogenous cell group** (동형세포집단)  
= aggregation of chondrocyte

Lacuna (연골소강)

Chondrocyte  
(연골세포)

**Interterritorial matrix**  
(영역사이기질)

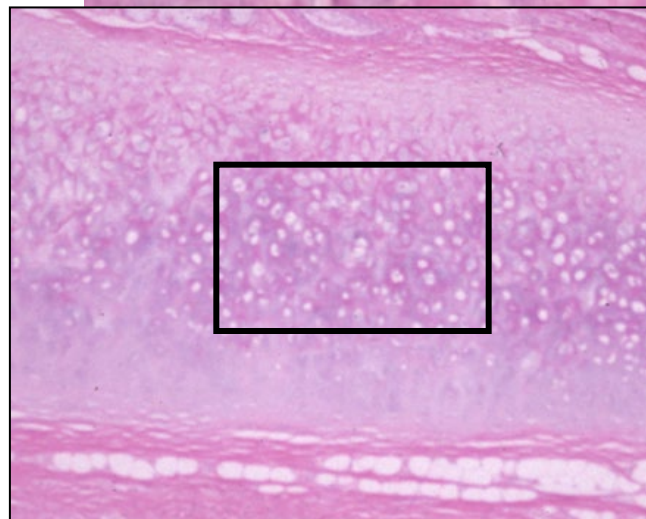
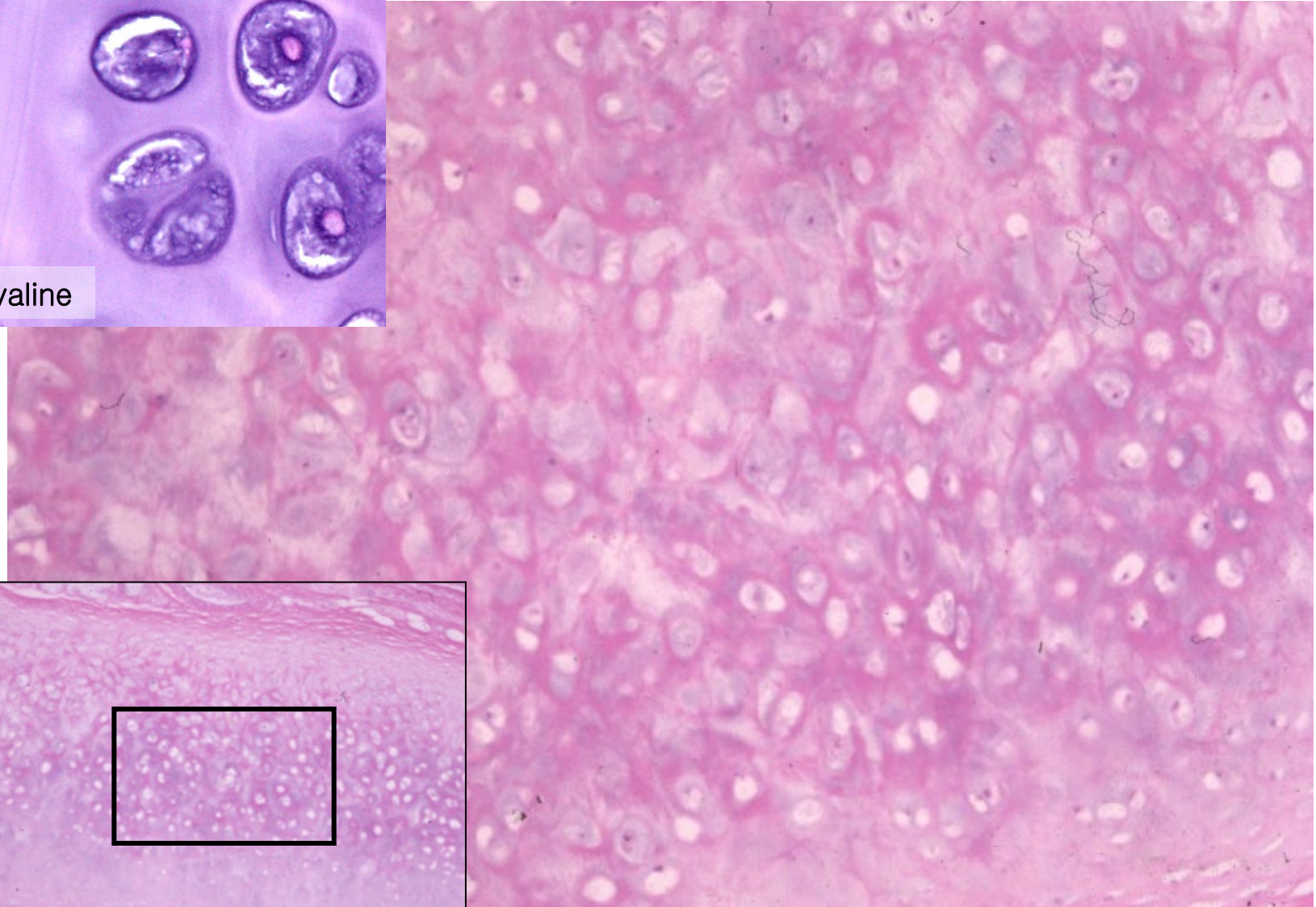
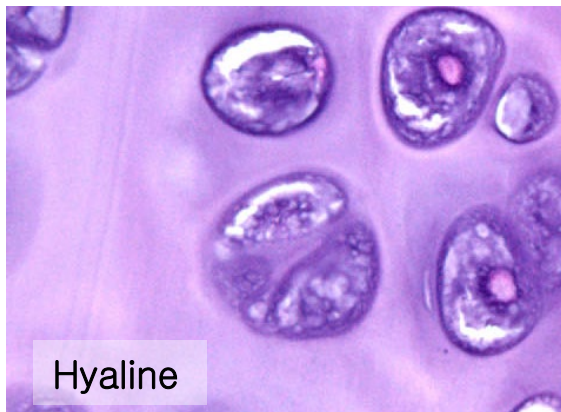


perichondrium

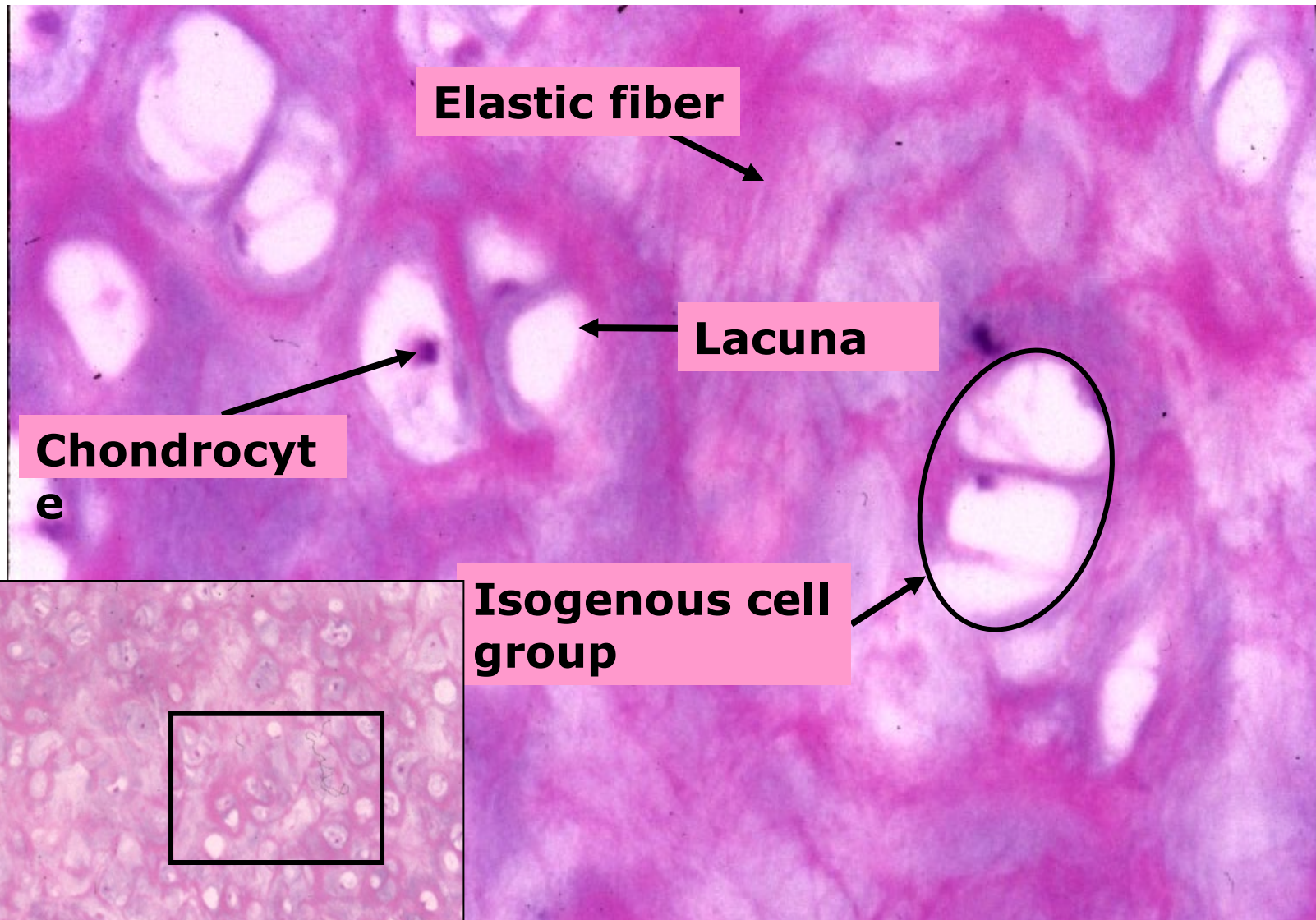
Epiglottic  
cartilage

perichondrium

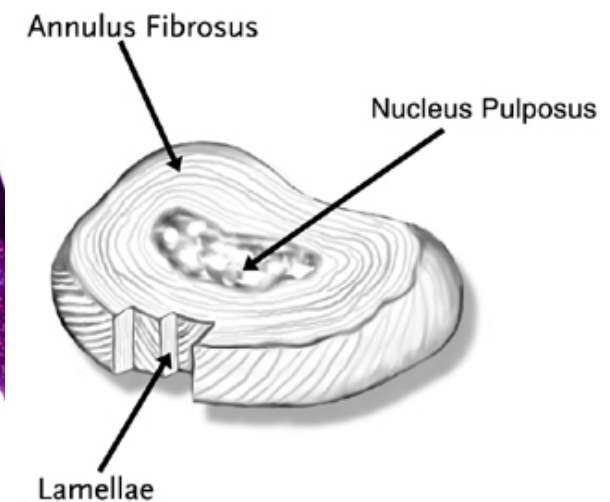
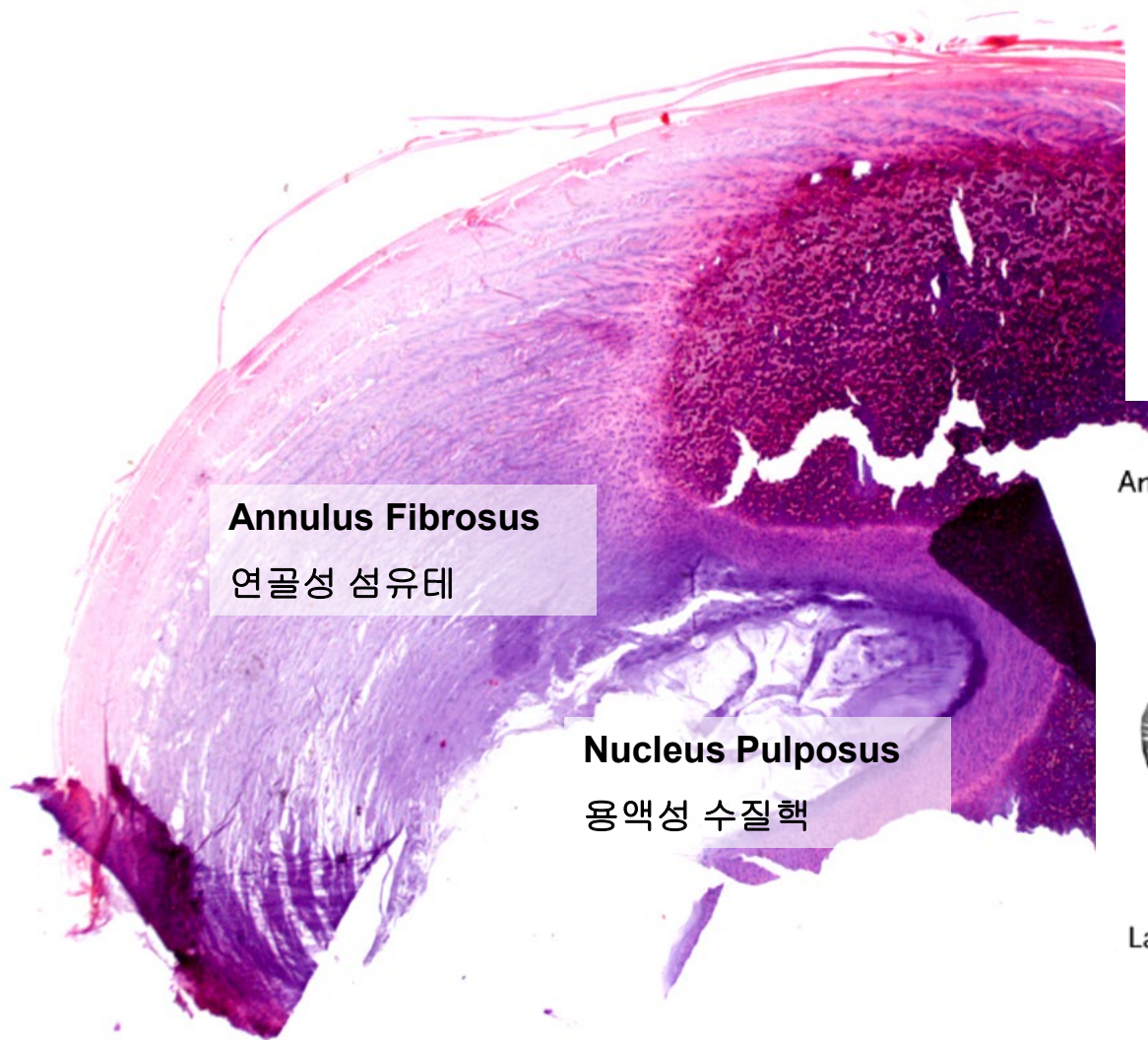
x40



x100

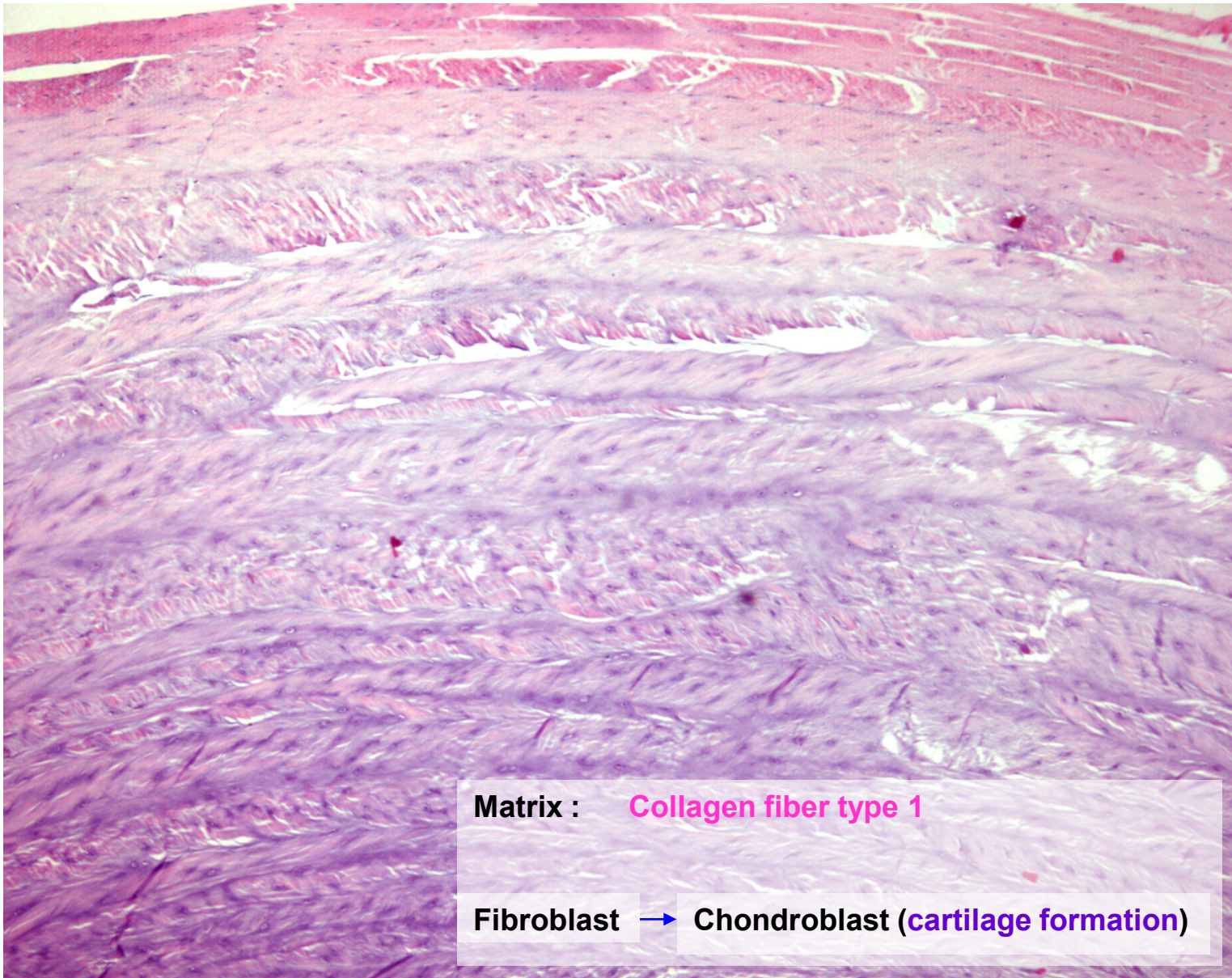


# #5 Intervertebral disc, human, HE, Fibrocartilage



**Stereomicroscopic view**

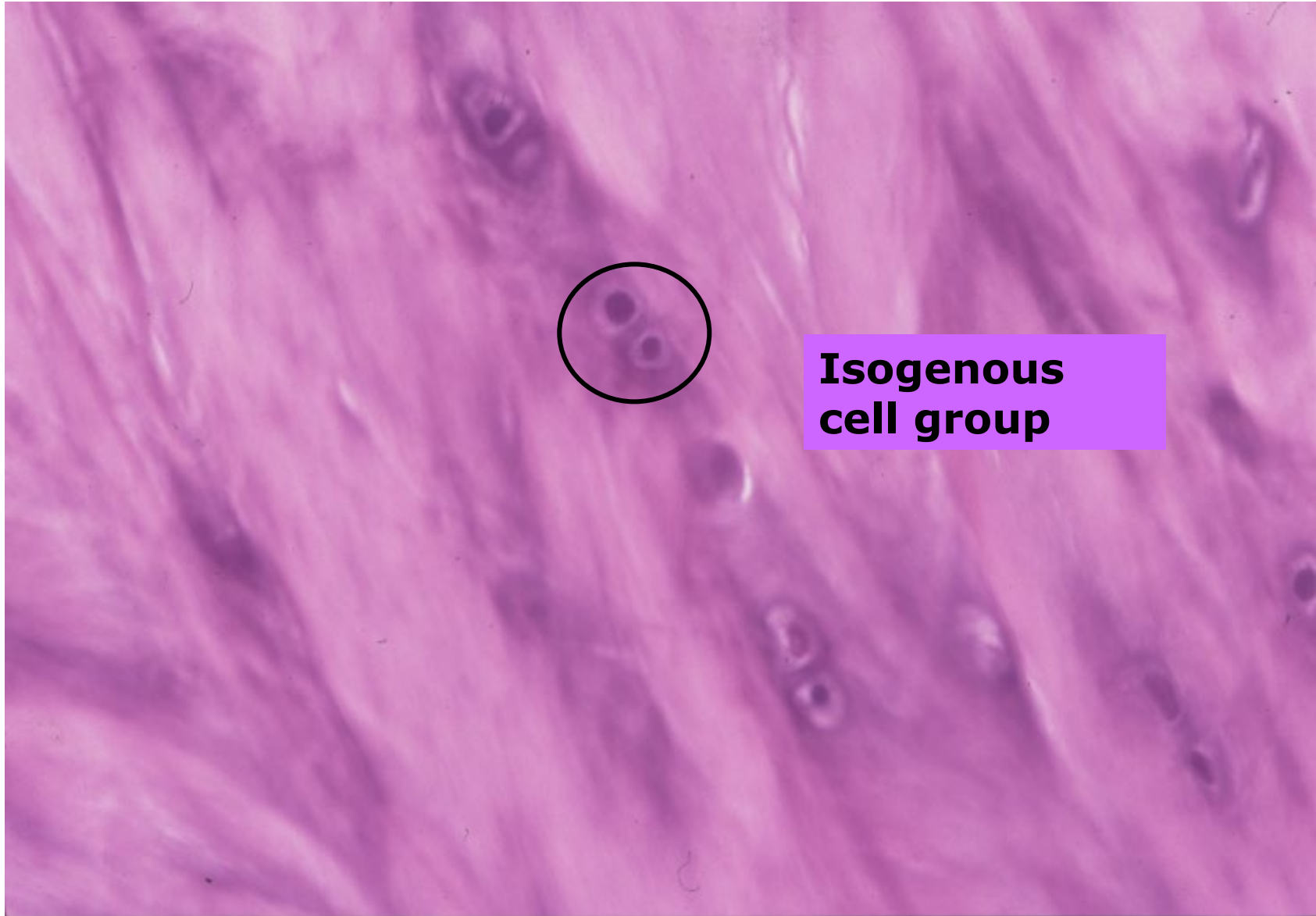
## #5 Intervertebral disc, human, HE, Fibrocartilage (섬유연골)



x40



**#5 Intervertebral disc, human, HE, Fibrocartilage**



**x400**

# Bone (뼈)

- ECM에 무기질이 많은 특수결합조직
- 골격형성, 장기보호, calcium농도 조절, bone marrow
- 구성
  - 1) Osteoblast(뼈모세포) : 뼈 기질성분 합성
  - 2) Osteocyte(뼈세포) : 기질에 둘러싸여 합성능력이 없어진 세포
  - 3) Osteoclast(뼈파괴세포) : 뼈의 용해와 흡수
  - 4) Bone matrix(뼈기질) : 무기질 + I형 collagen + 무형질
- 유형
  - 1) Compact bone (치밀뼈)
  - 2) Cancellous or spongy bone (해면뼈)
- Decalcification(탈회) 필요

**#3 Bone (decalcified), HE, human, ×40**

**#2 Bone (cross), ground, human, ×400**

**#OH26 Bone development, HE, human, ×400**

**Intramembranous bone formation**

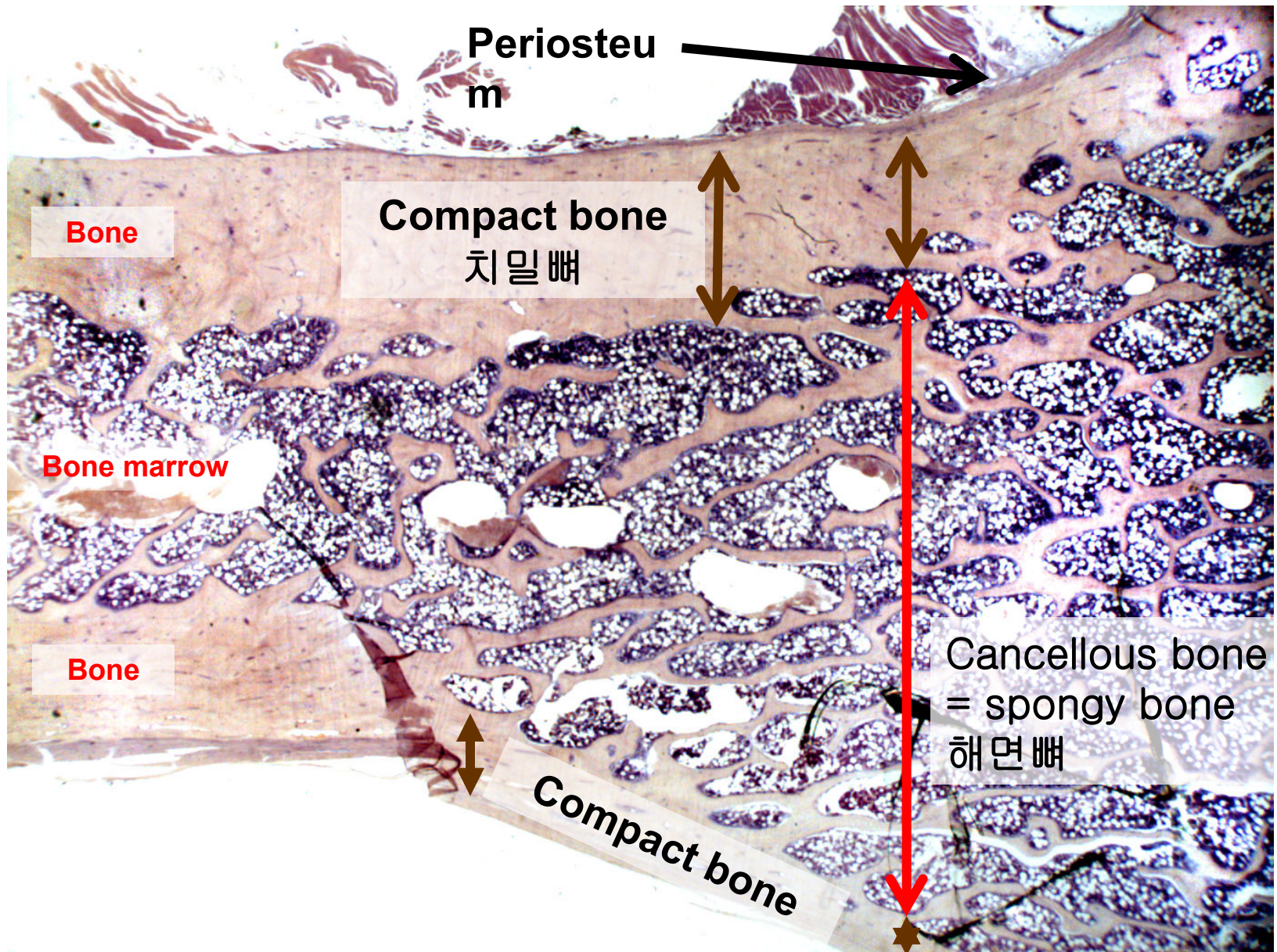
**#80 Bone development, Mallory, human, ×400**

**Endochondral bone formation**

**#4 Bone development, HE, human**

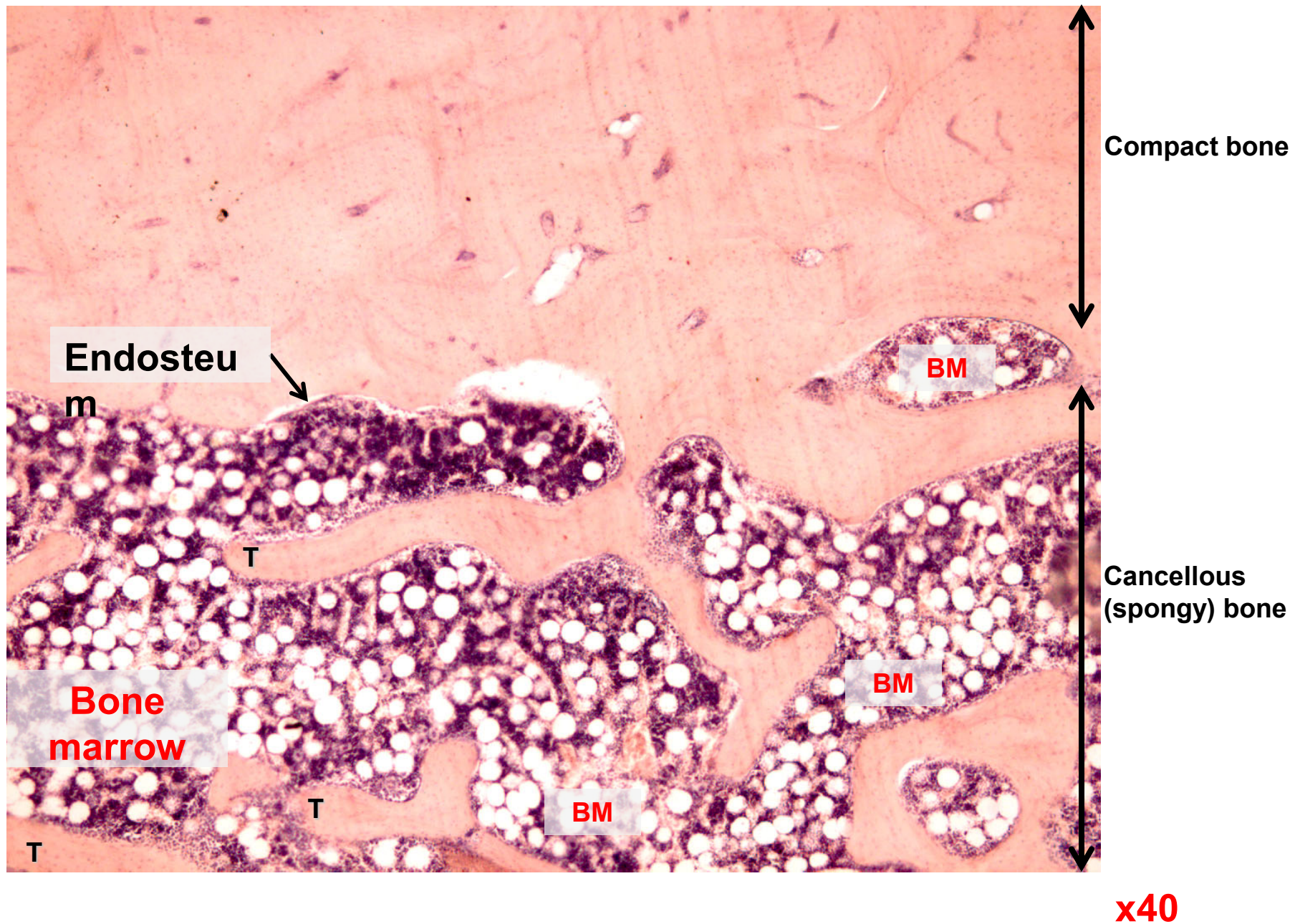
**Endochondral bone formation**

#3 Bone (decalcified with acidic soln.), human, HE,

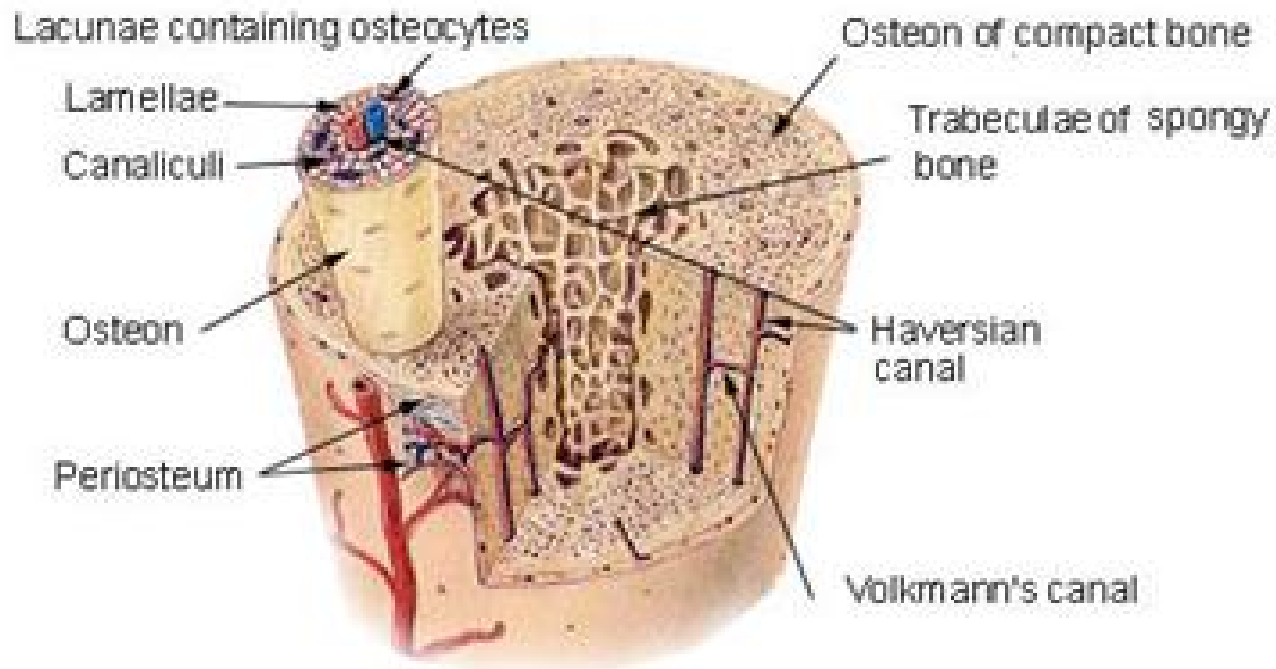
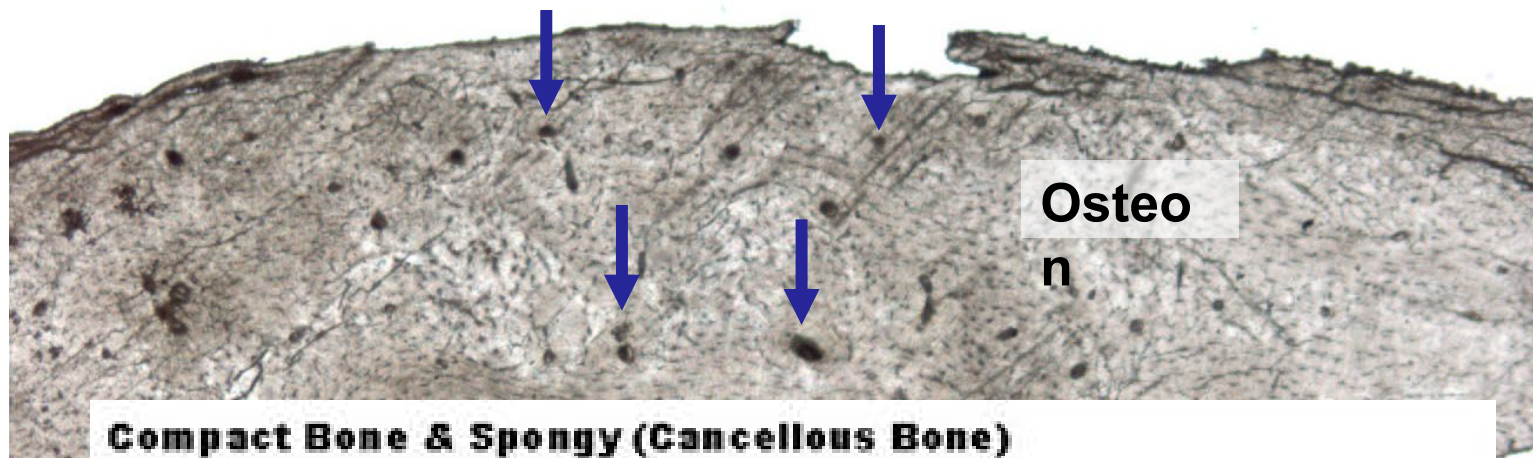


**Stereomicroscopic view**

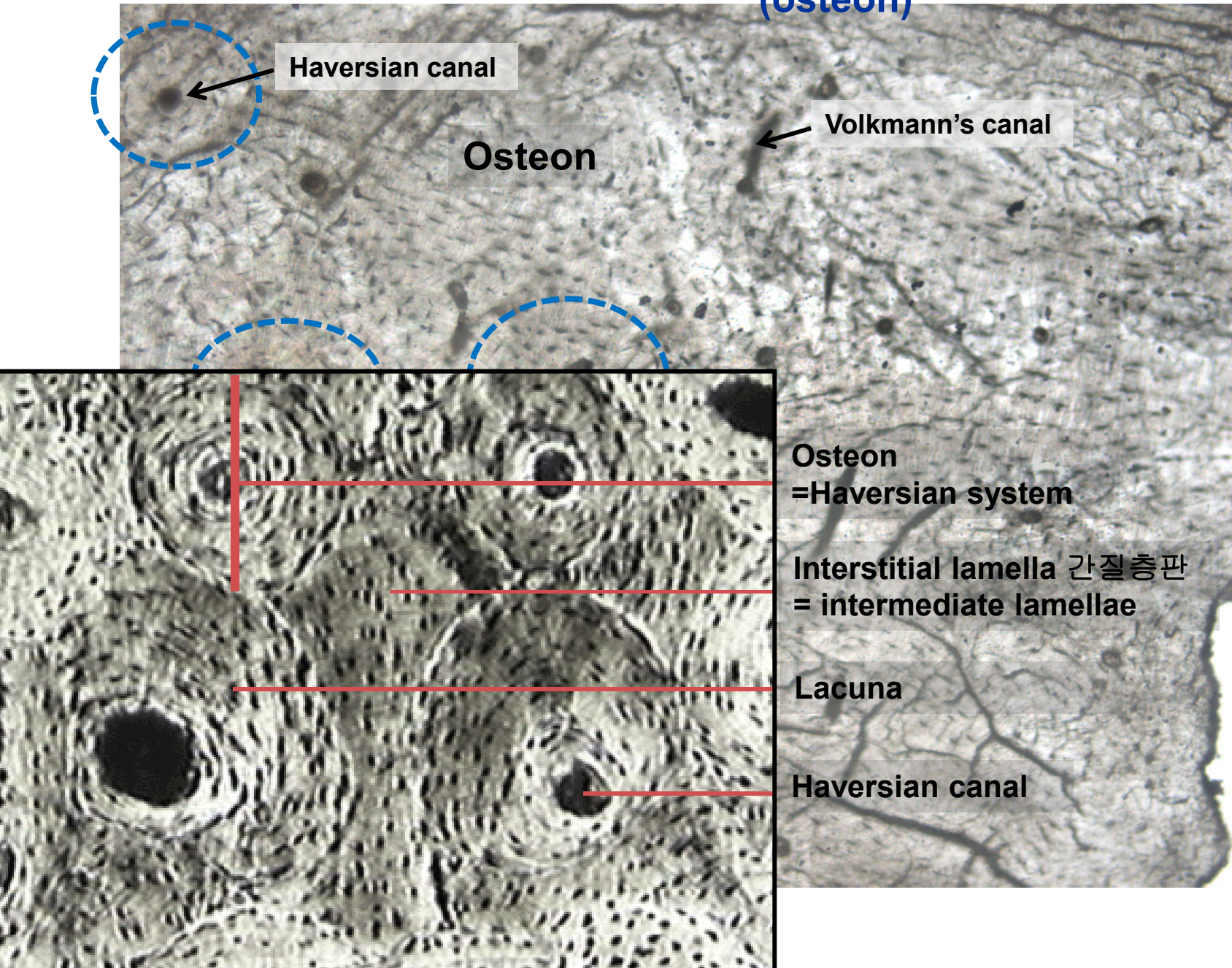
**#3 Bone (decalcified with acidic soln.), human, HE,**



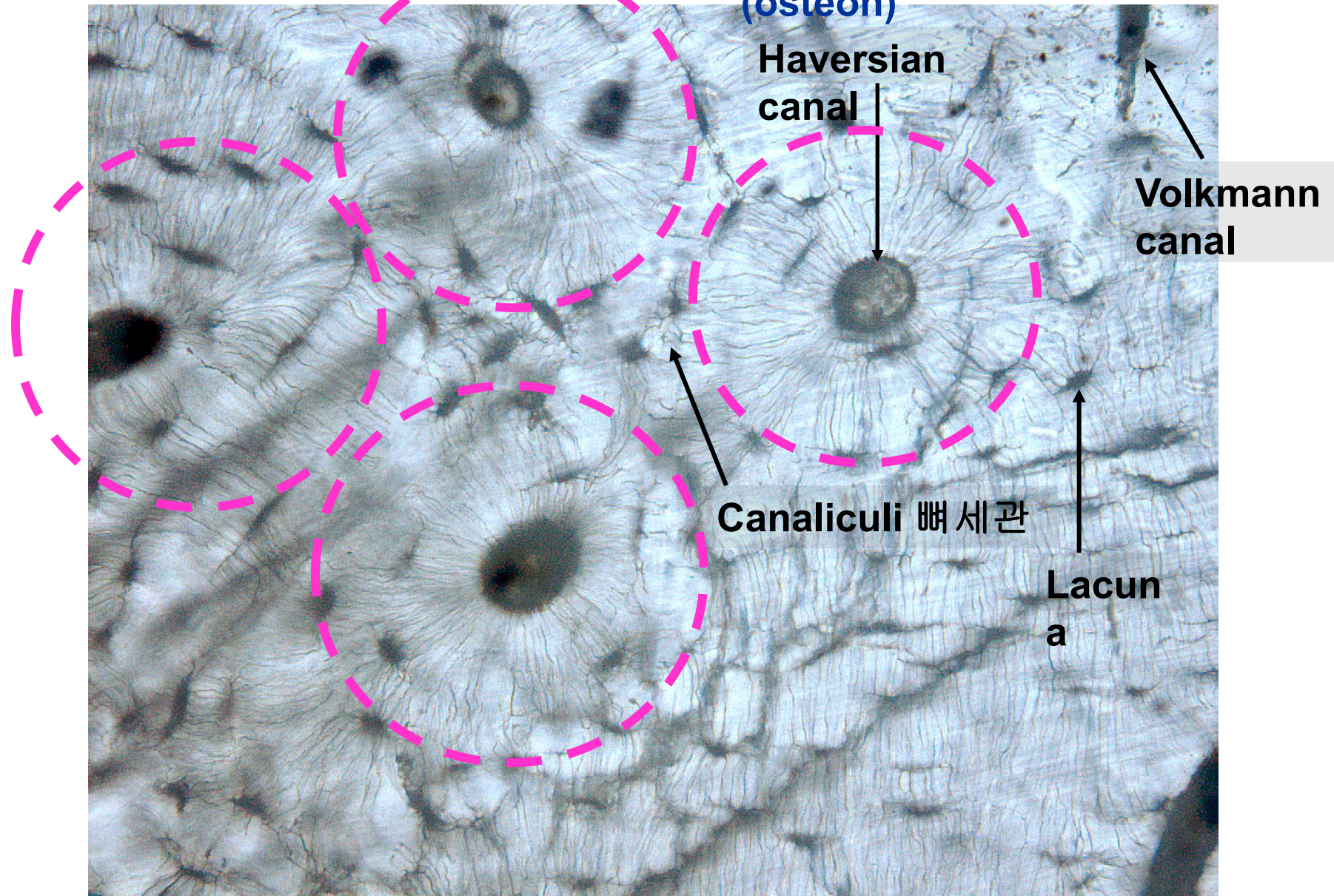
## #2 Bone (ground preparation), human, Haversian system (osteon)



## #2 Bone (ground preparation), human, Haversian system (osteon)

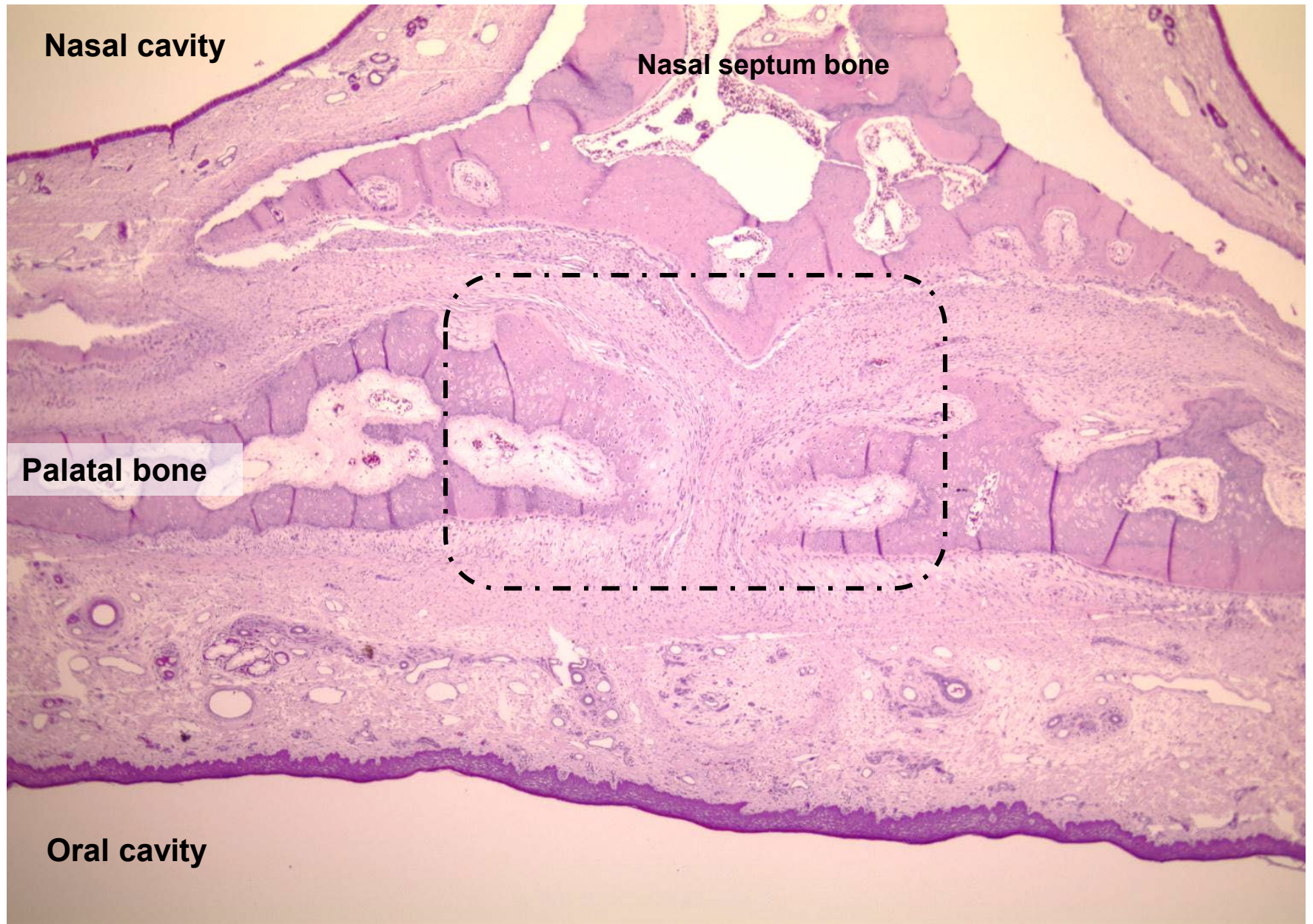


## #2 Bone (ground preparation), human, Haversian system (osteon)



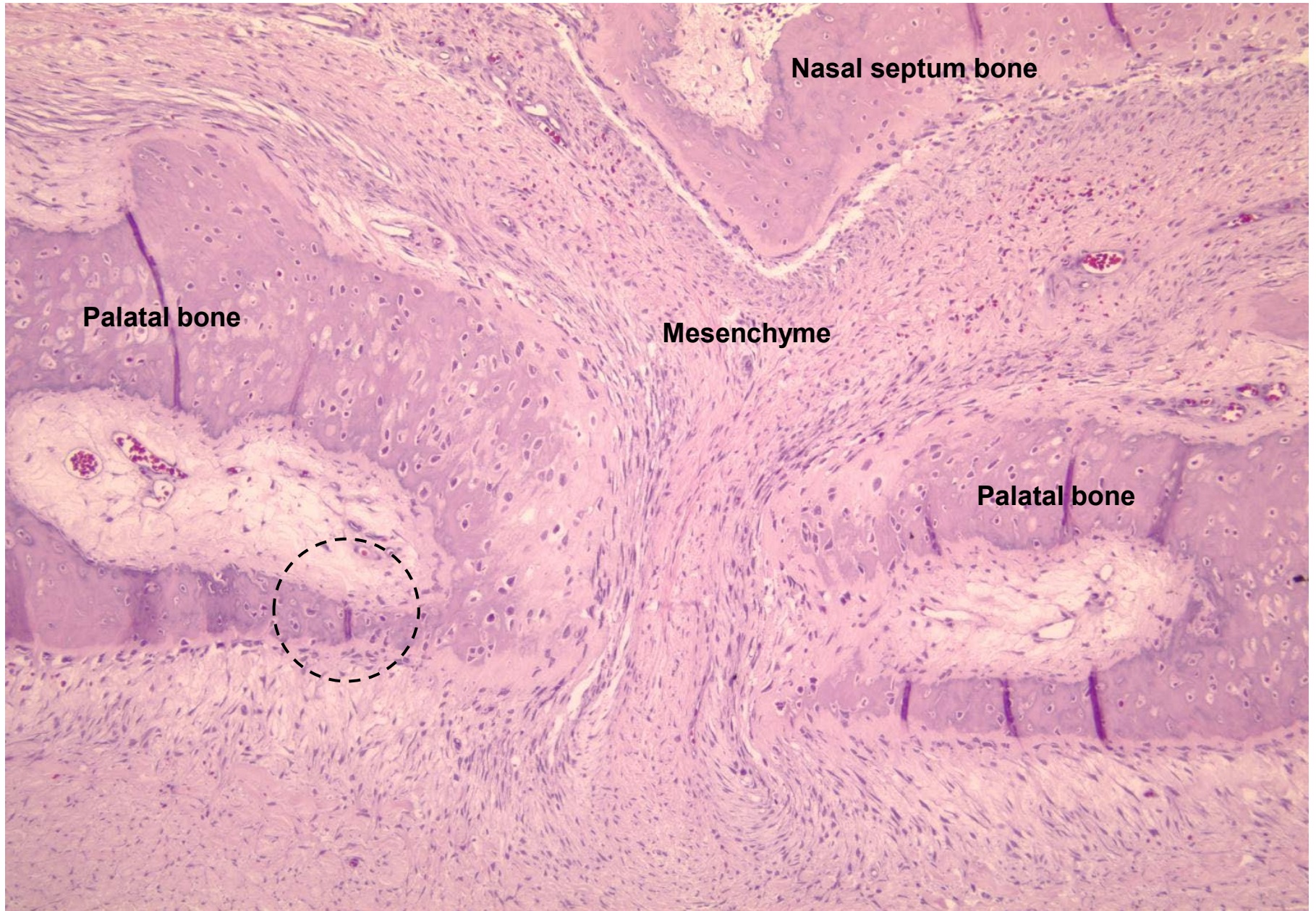
x400

# #OH26 Bone development, human, HE, Intramembranous bone formation



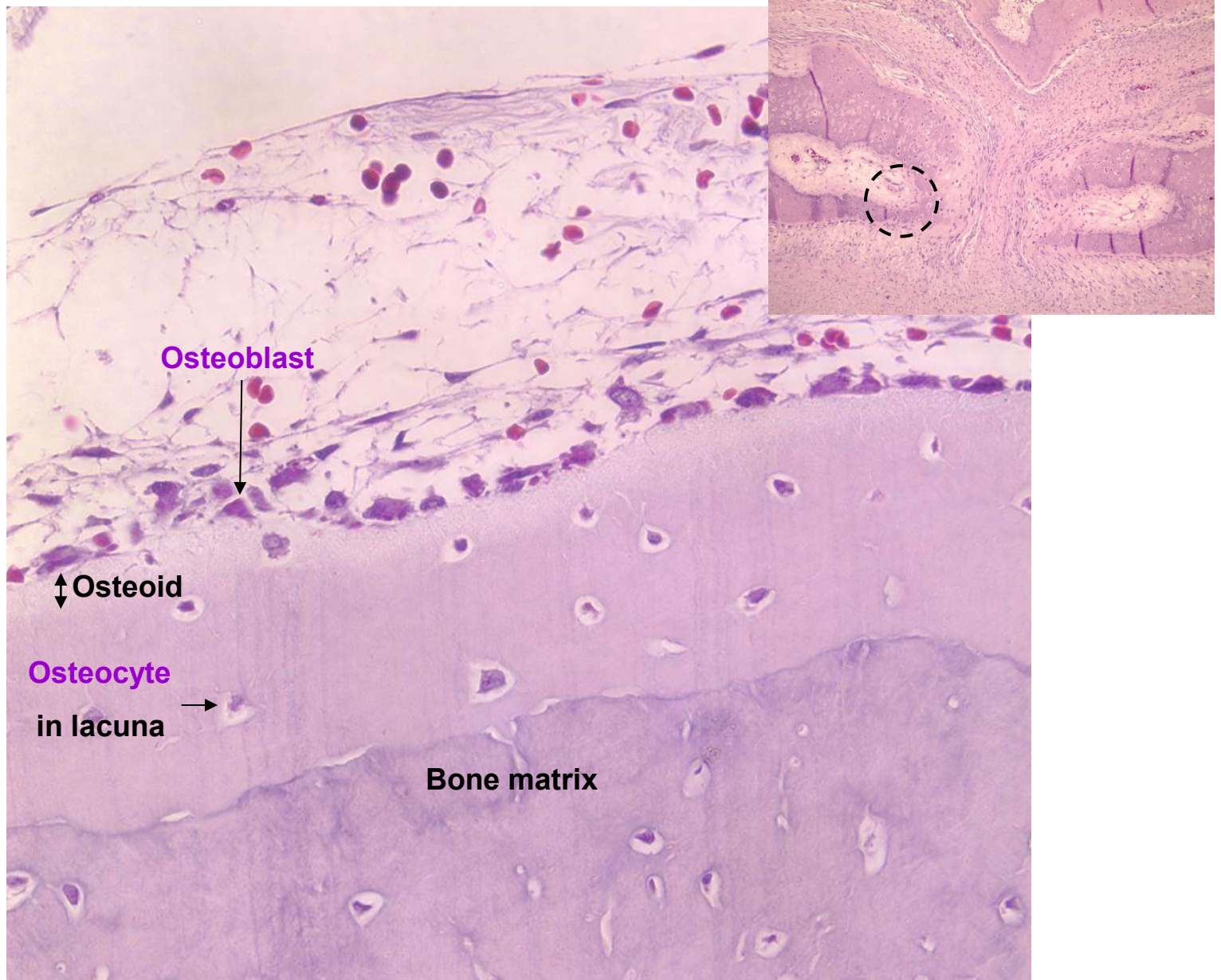
x40

# #OH26 Bone development, human, HE, Intramembranous bone formation

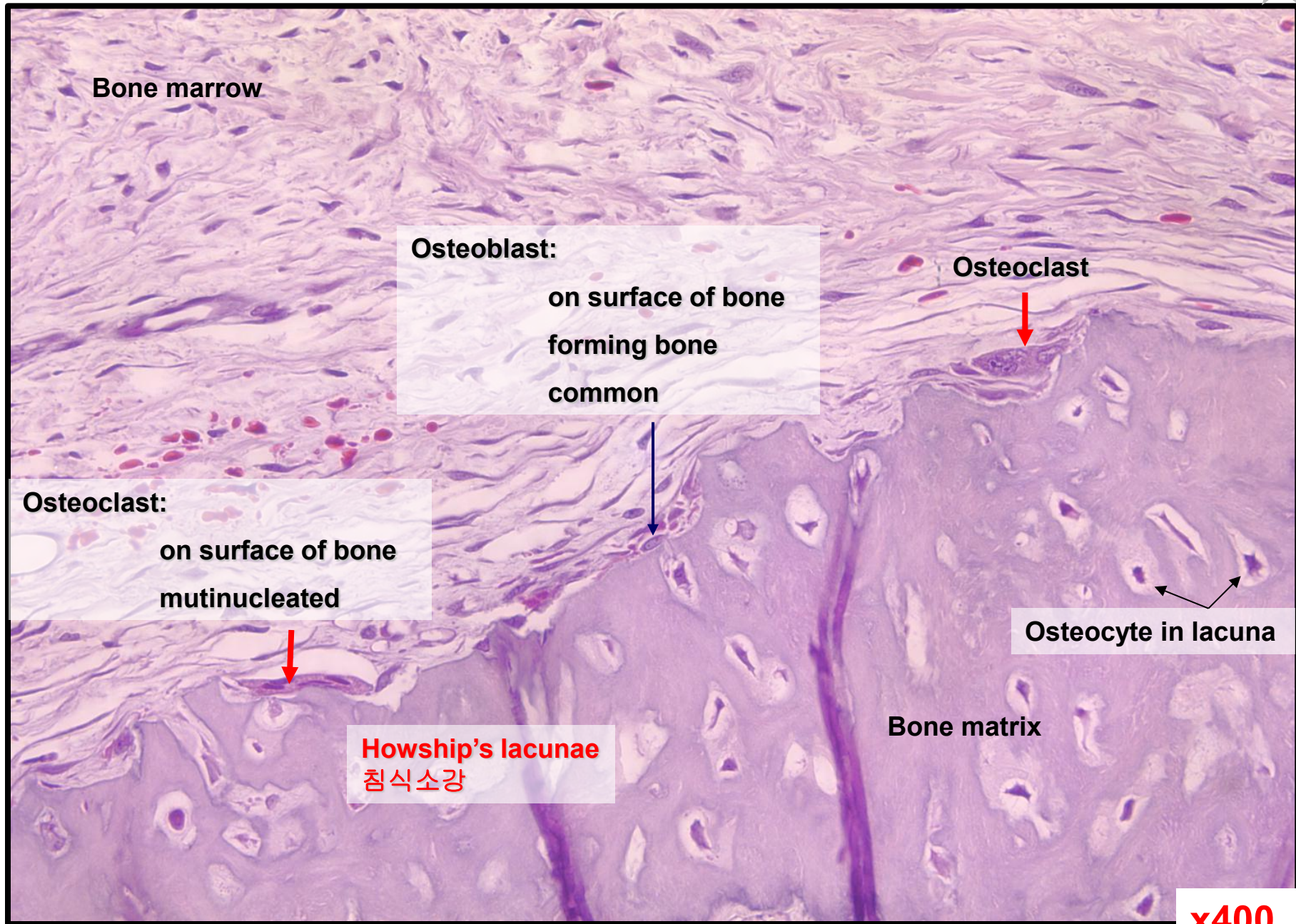


x100

# #OH26 Bone development, human, HE, Intramembranous ossification



x400



# #80 Bone development, human, Azan-Mallory, Endochondral bone formation

## Azan-Mallory stain

Red – Nuclei

Blue – Collagen

Light blue – Cartilage

Bone- Blue (early)

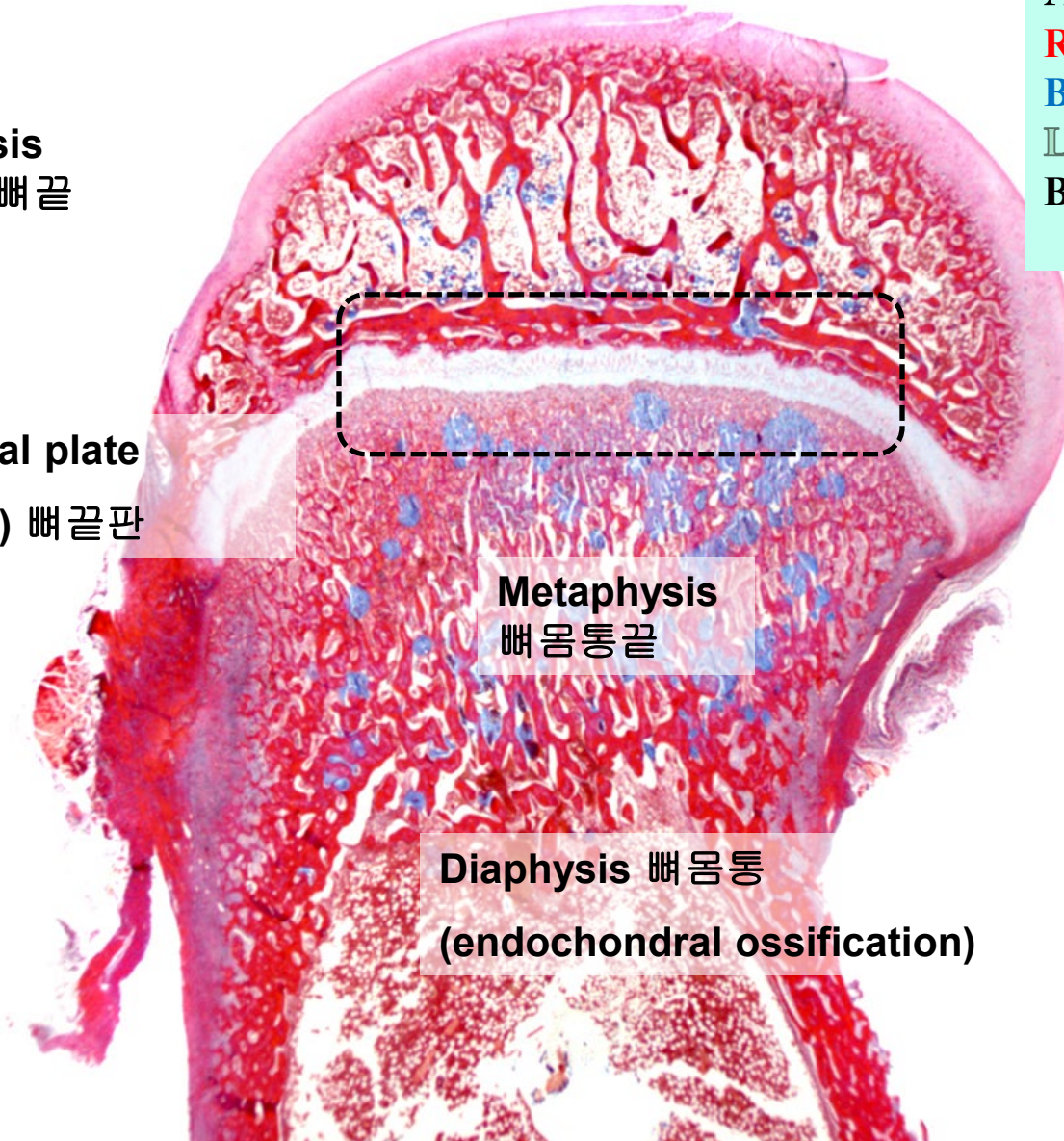
Red (mature)

Epiphysis  
(Bone) 뼈끝

Epiphyseal plate  
(cartilage) 뼈끝판

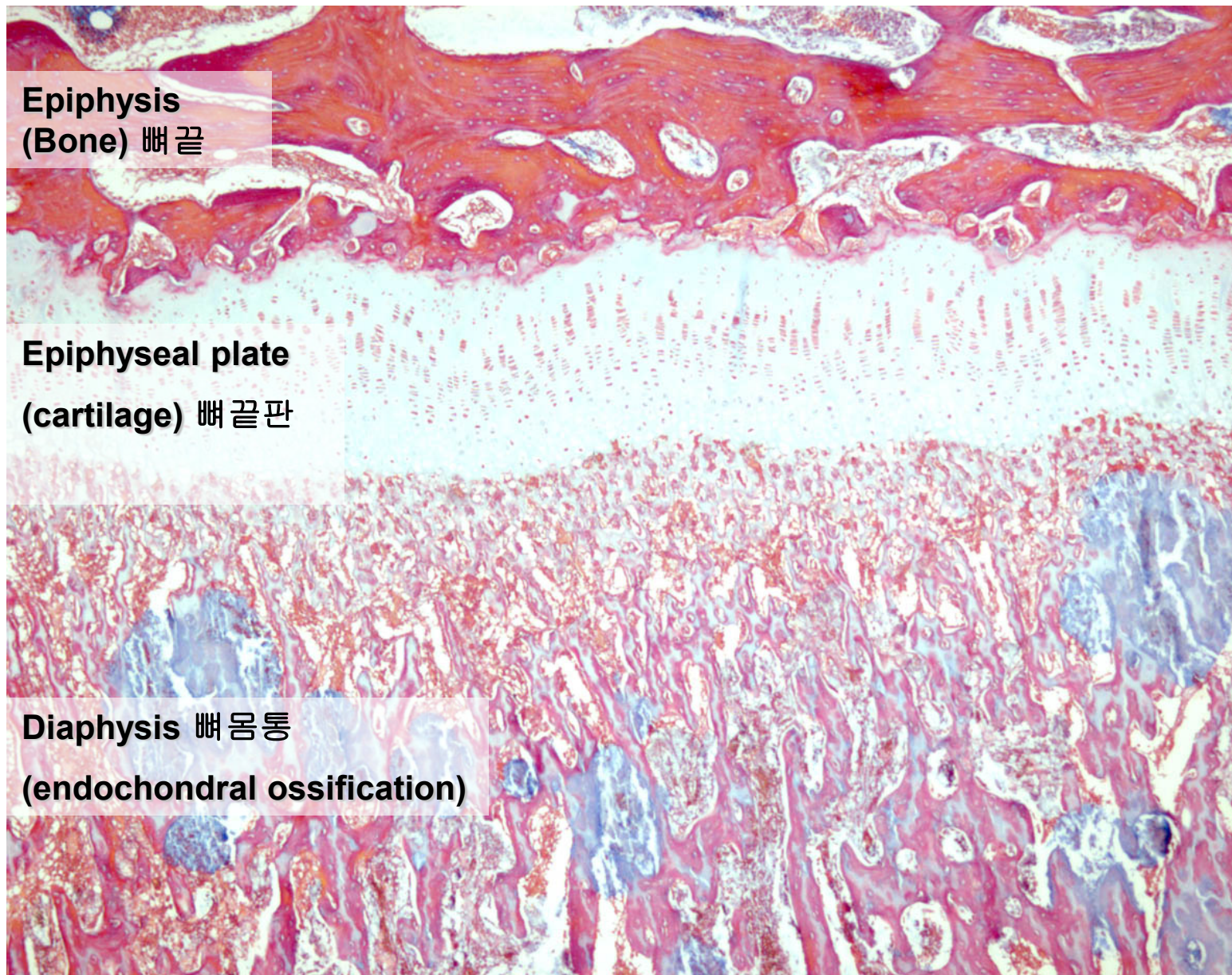
Metaphysis  
뼈몸통끝

Diaphysis 뼈몸통  
(endochondral ossification)



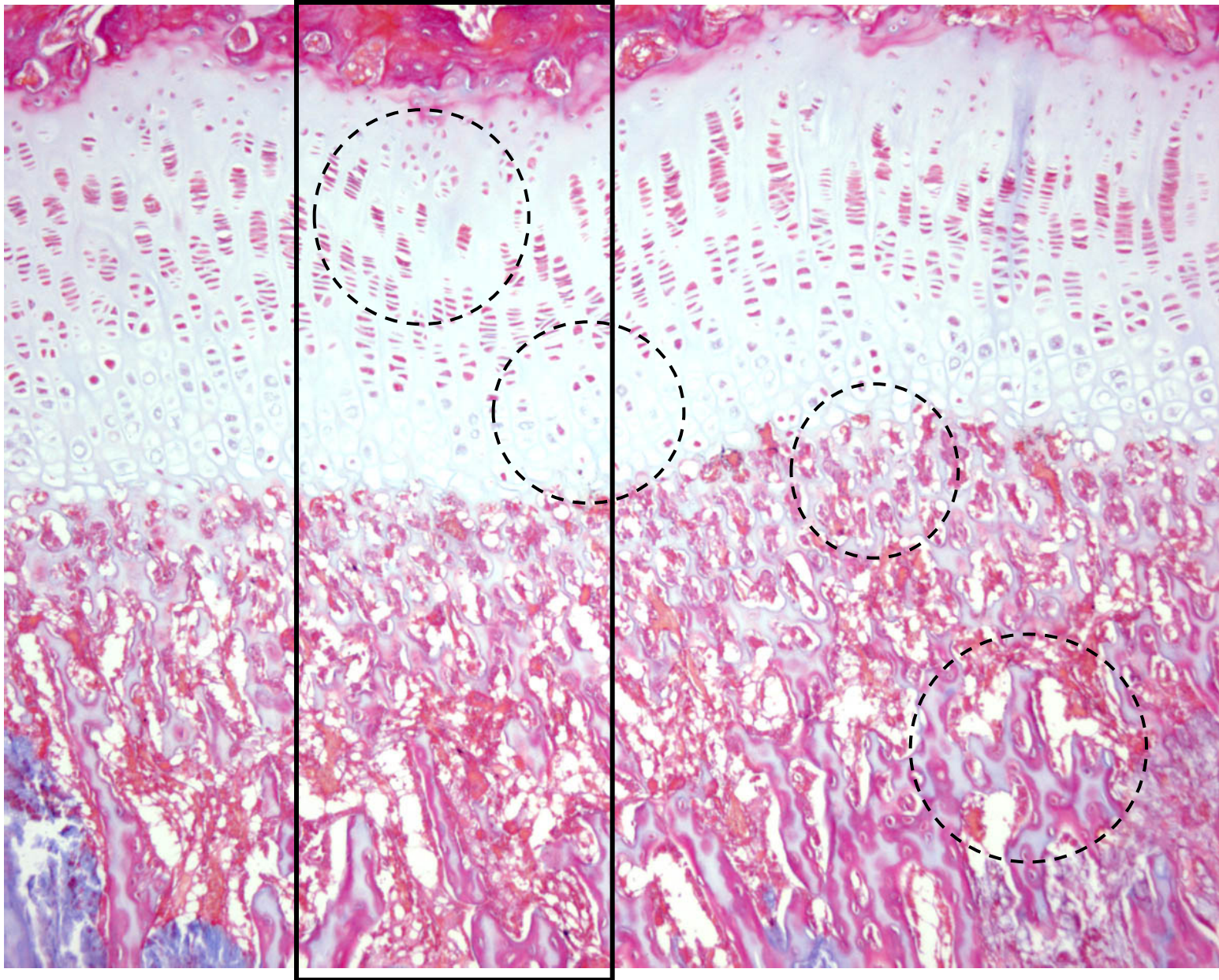
Stereomicroscopic view

## #80 Bone development, human, Mallory, Endochondral ossification



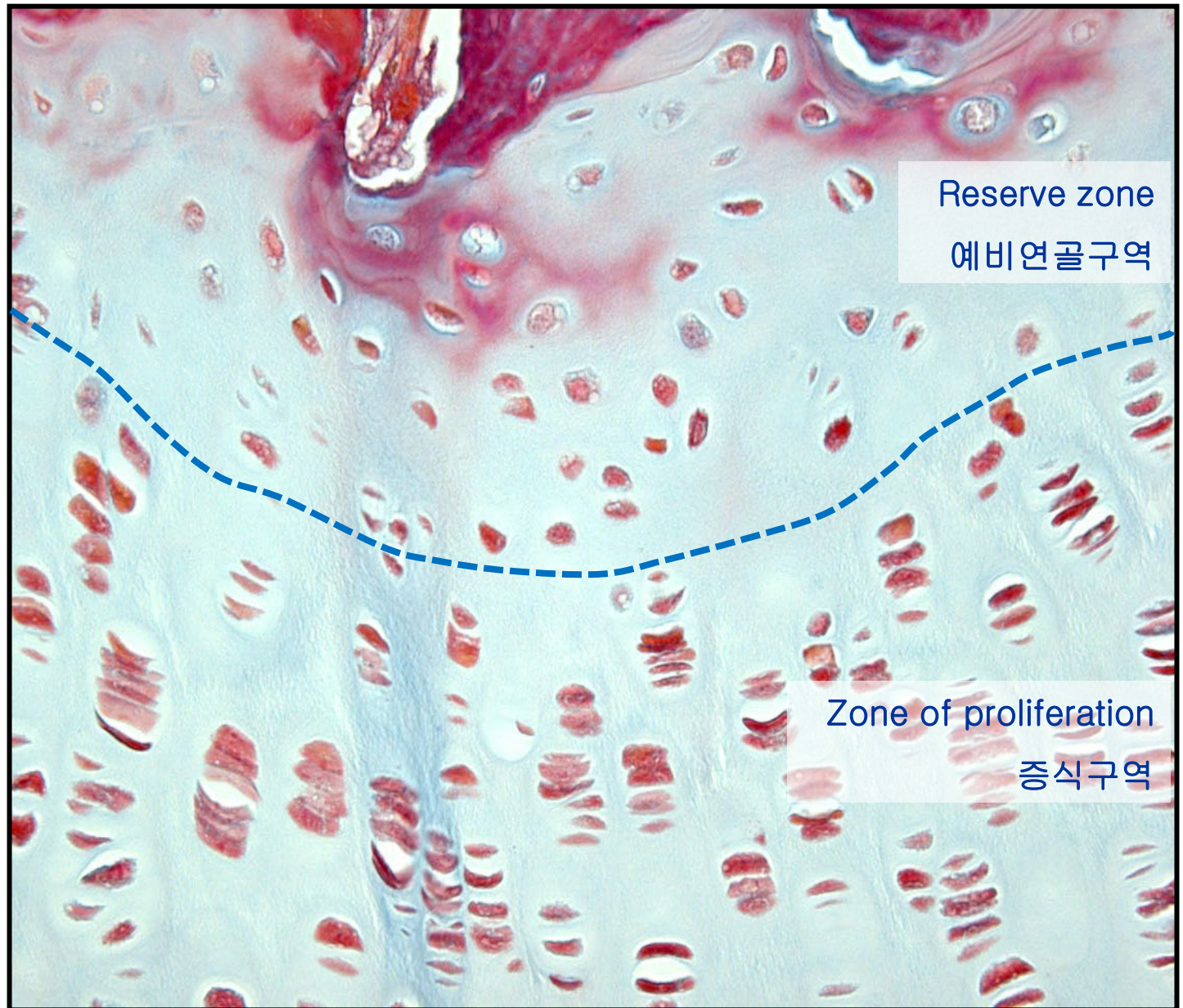
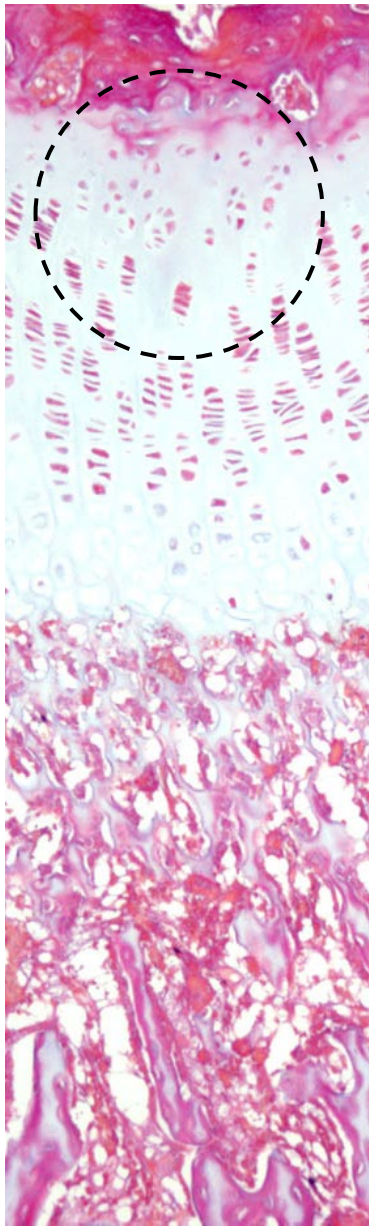
x50

# #80 Bone development, human, Azan-Mallory, Endochondral bone formation



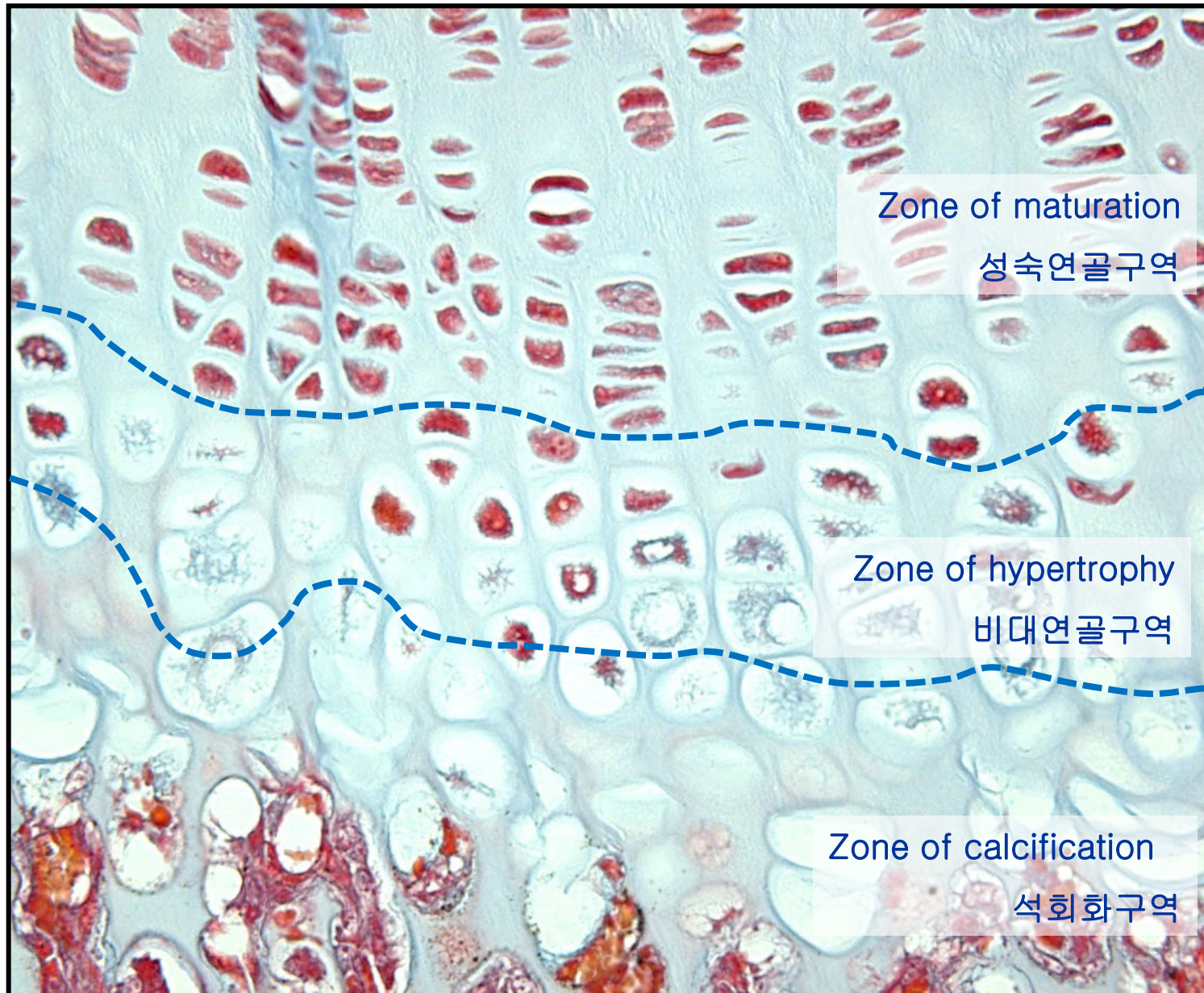
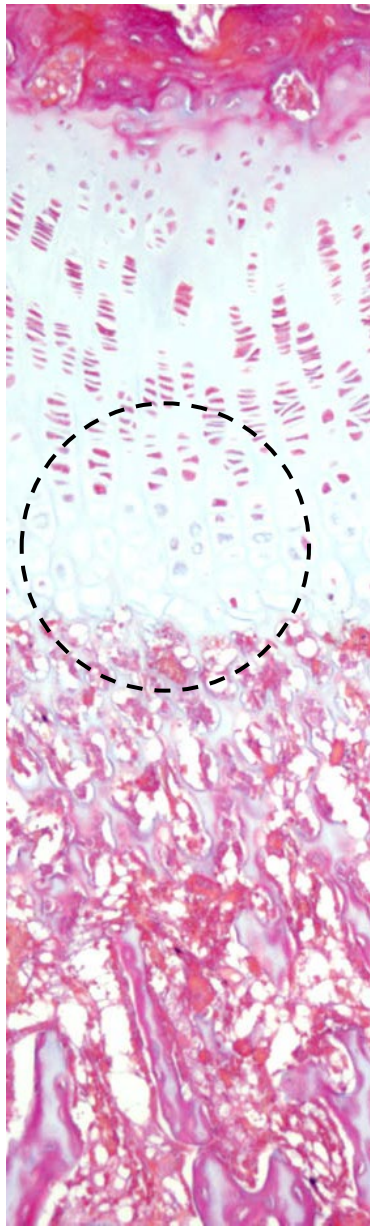
x100

# #80 Bone development, human, Azan-Mallory, Endochondral bone formation



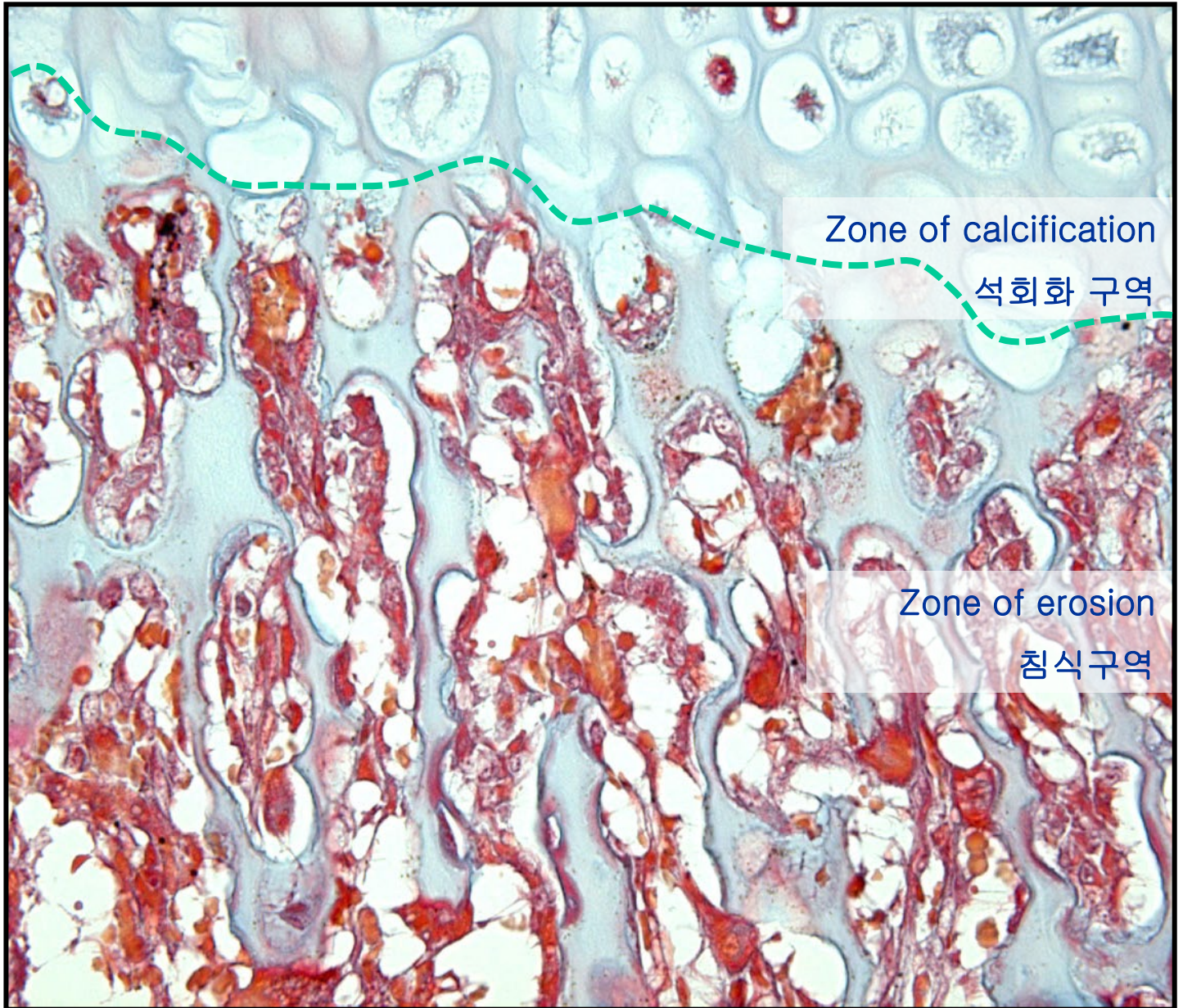
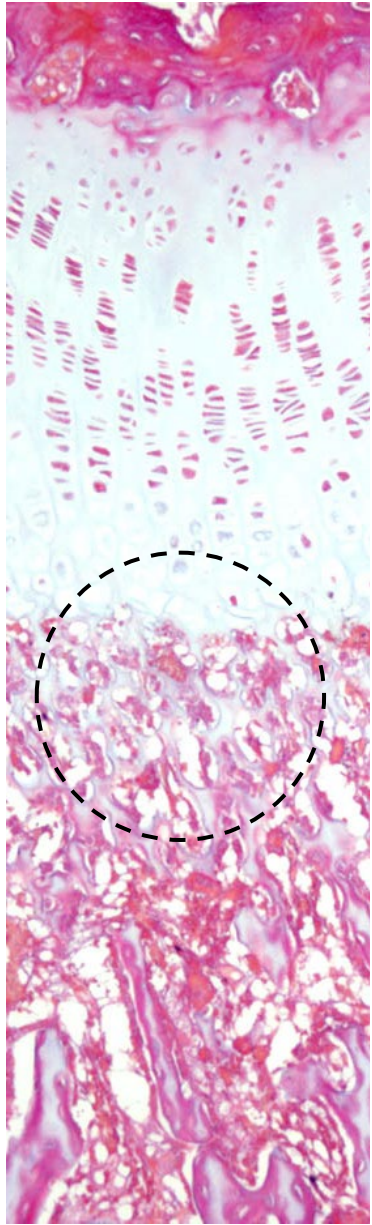
x400

# #80 Bone development, human, Mallory, Endochondral ossification



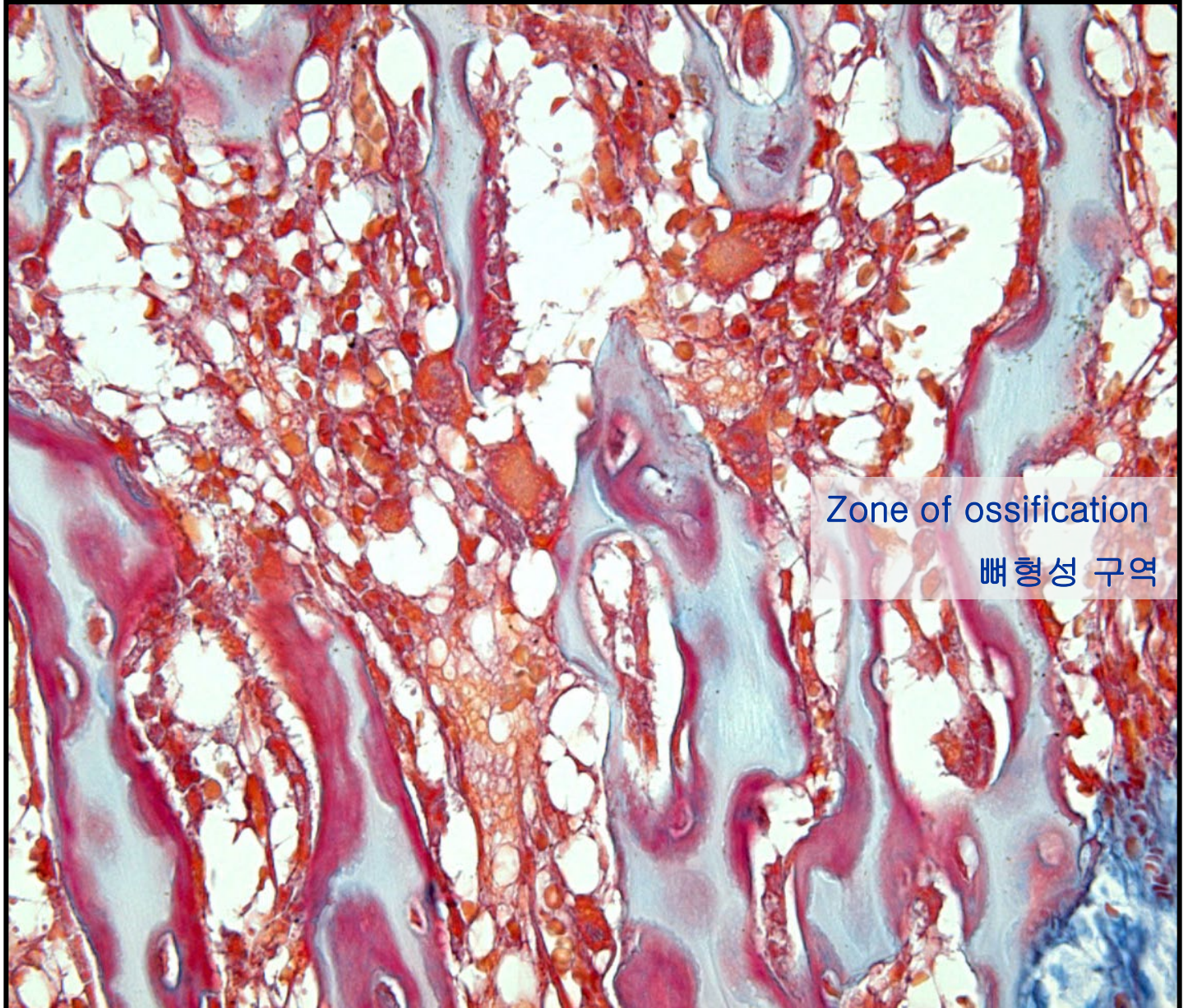
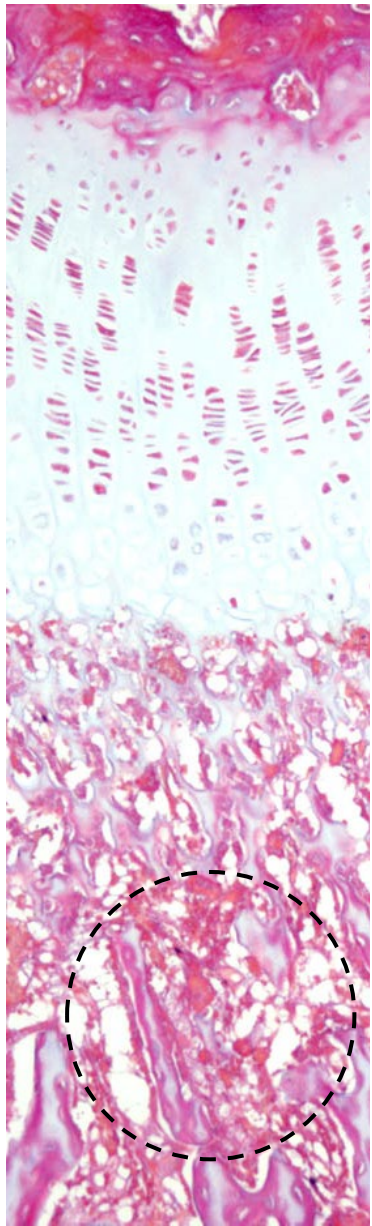
x400

# #80 Bone development, human, Azan-Mallory, Endochondral bone formation



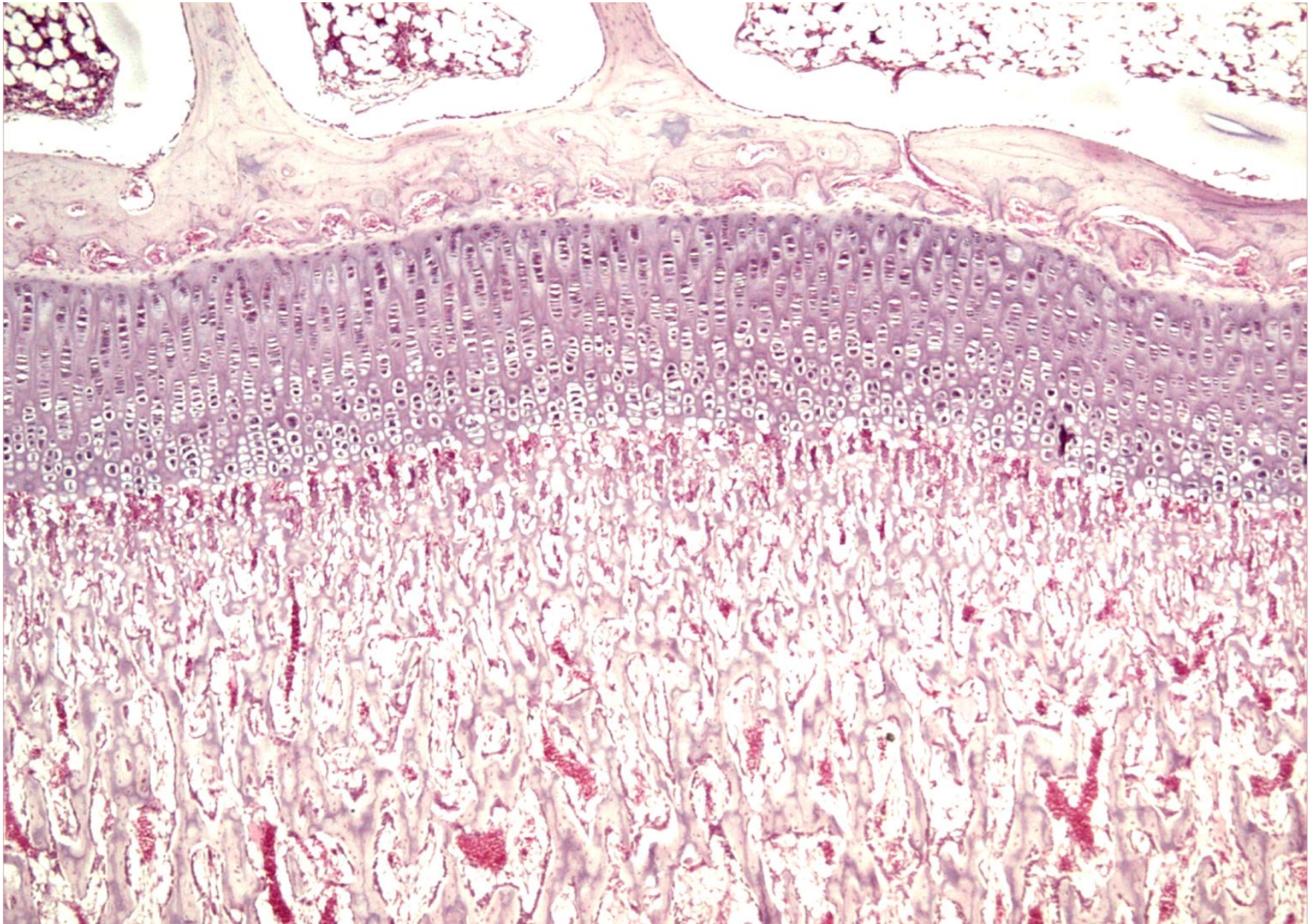
x400

# #80 Bone development, human, Azan-Mallory, Endochondral bone formation



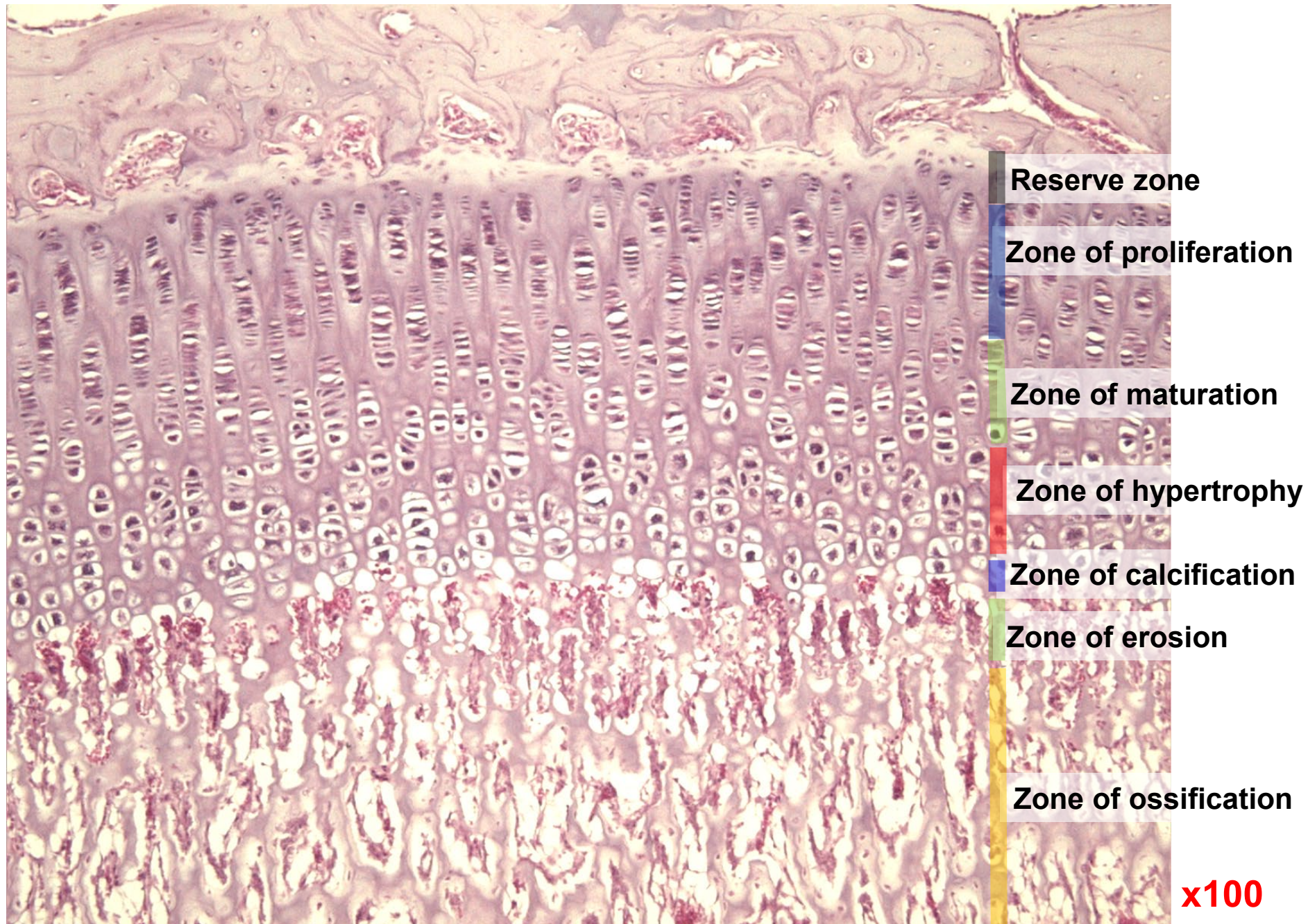
x400

#### #4 Bone development, human, H&E, Endochondral bone formation



x40

## #4 Bone development, human, H&E, Endochondral bone formation



# #4 Bone development, human, H&E, Endochondral bone formation

Reserve zone

예비연골 구역

Zone of proliferation

증식 구역

Zone of maturation

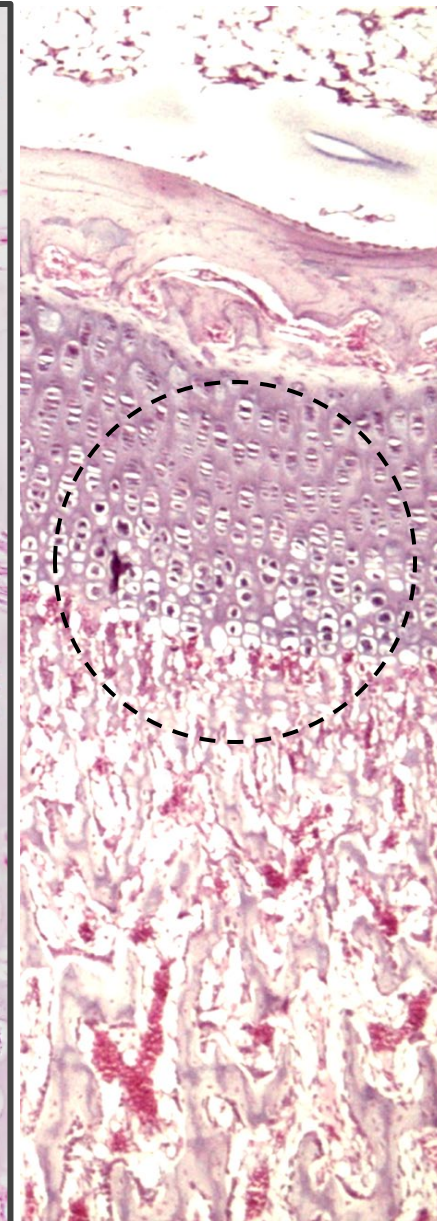
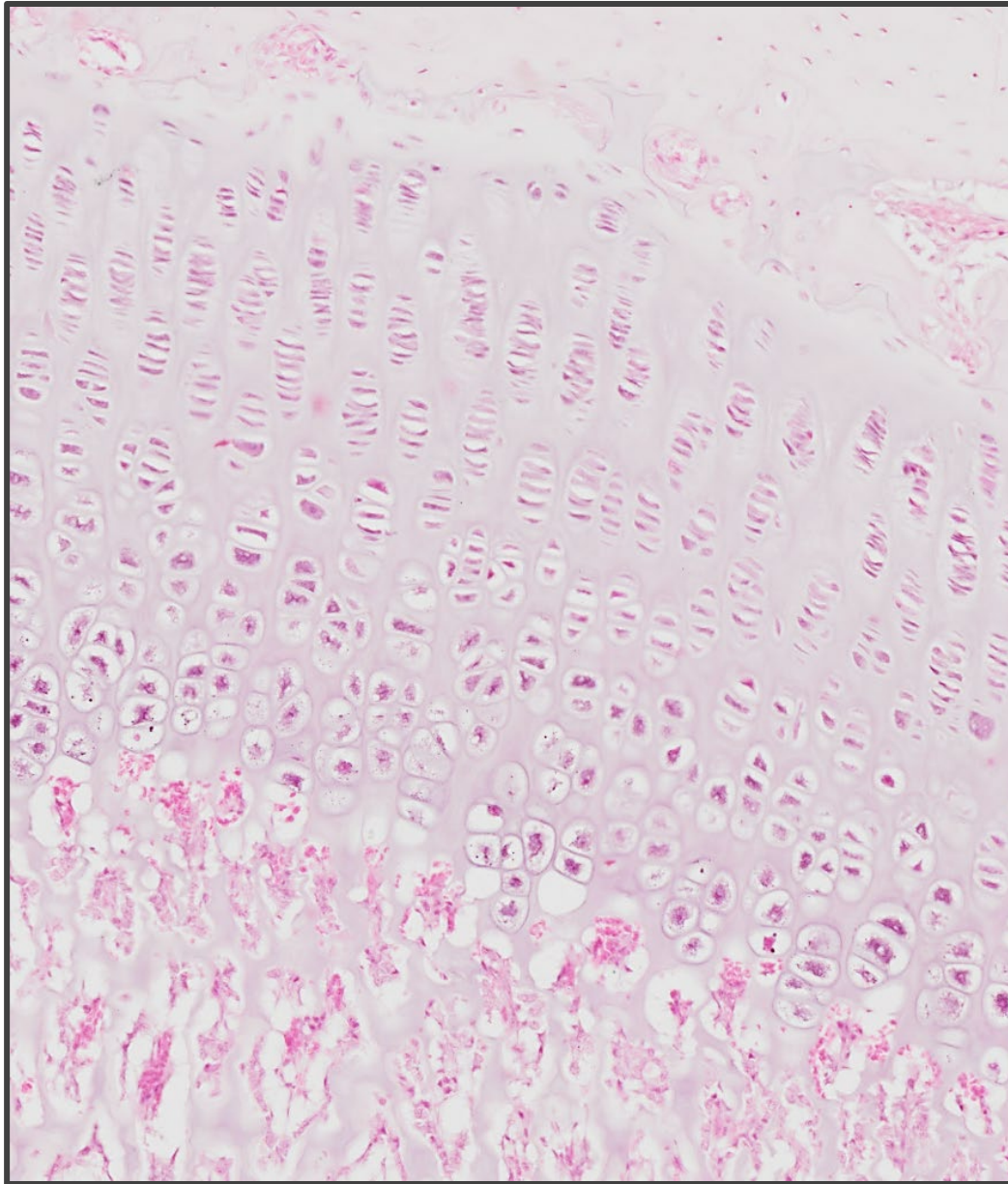
성숙연골 구역

Zone of hypertrophy

비대연골 구역

Zone of calcification

석회화 구역



x400

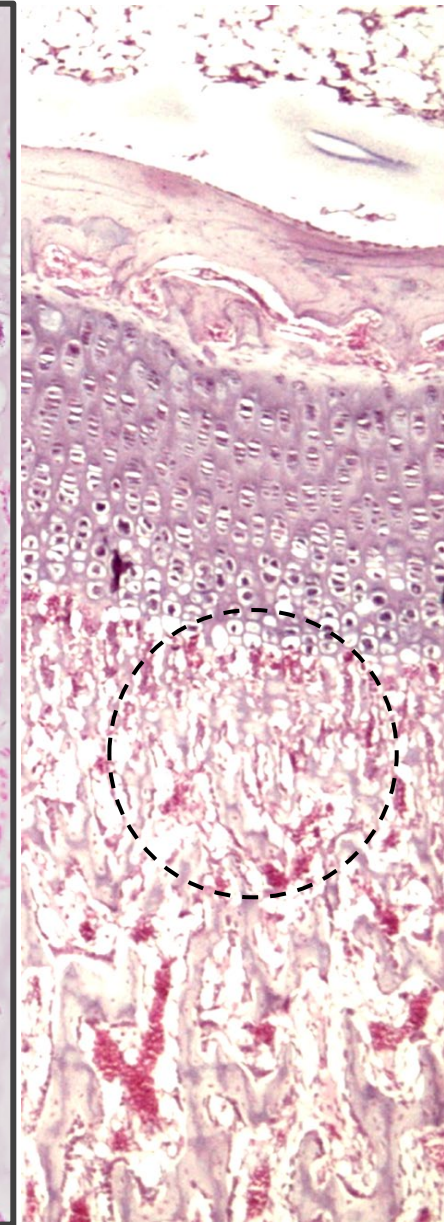
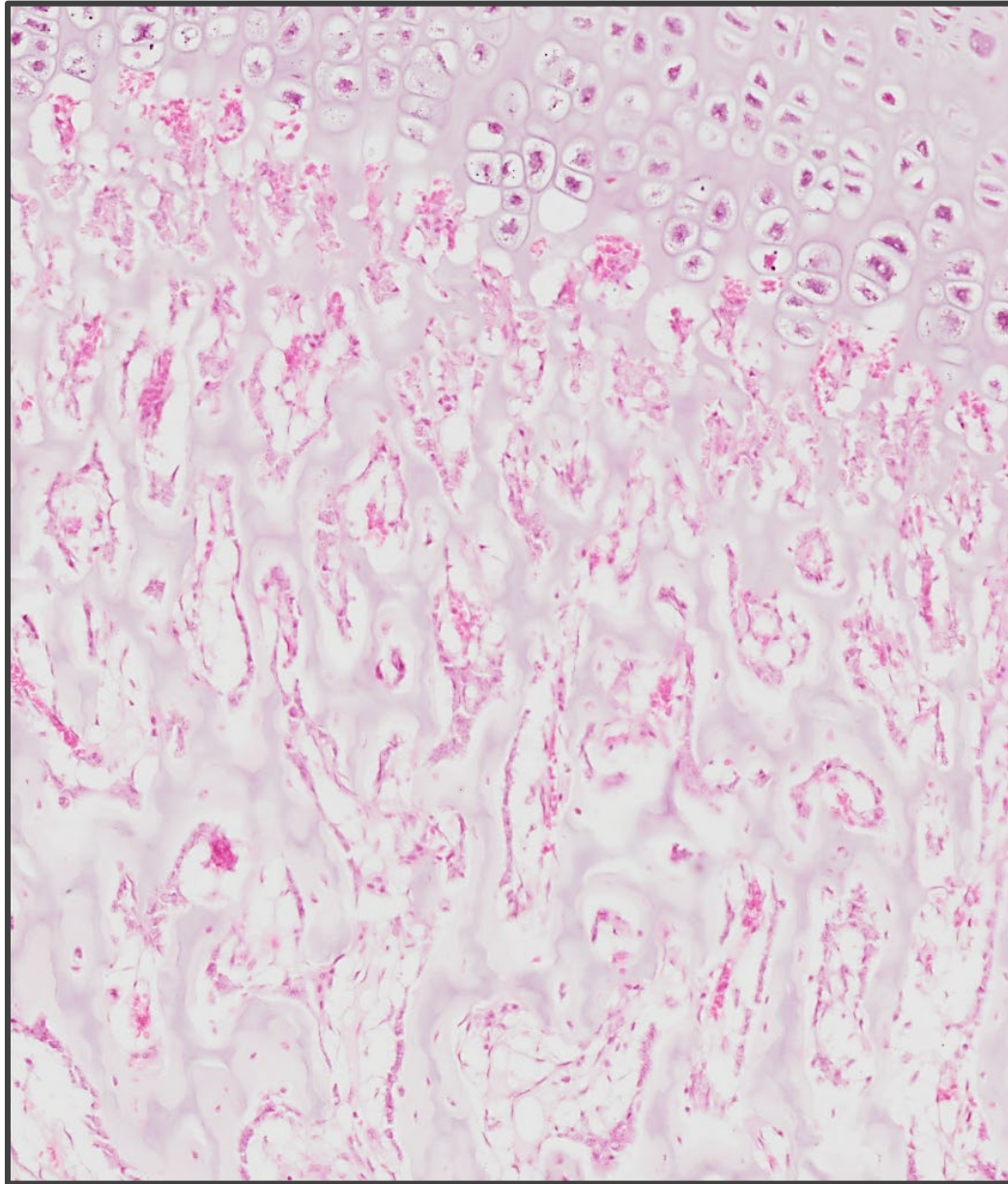
## #4 Bone development, human, H&E, Endochondral bone formation

Zone of erosion

침식 구역

Zone of ossification

골화 구역



x400